Solid States

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Solid States

Solid state has a huge contribution in infrastructure and growth of society. Right from the beginning, us humans have been dependent on solids for survival. From tools made for hunting to houses made for living even our ancestors used solid materials everywhere. Even in modern world we use solids for different infrastructural and technological developments. From semiconductors used in our cell phones to the parts of a modern Tesla car we can observe solids everywhere.

In this chapter we will discuss mostly about how at microscopic level solids can be studied and distinguished based on properties and their molecular structures.

Matter can be classified, mainly in three categories based on their physical properties and molecular arrangements, solids, liquids, and gases. The table below includes properties of these states of matter which distinguish them from one another.

Properties	Solid	Liquid	Gas
shape	definite	Indefinite	Indefinite
volume	definite	definite	Indefinite
Interparticle forces	Strong	Weak to strong	negligible
Interparticle space	very least	least	large
Compression	least	least	Can be compressed
Effect of temp.	negligible	negligible	large of effect
Specific heat	constant	constant	variable
Inter particle forces	Attraction tendencies dominate over escaping tendencies	Attraction tendency weakly dominate over escaping tendencies	Escaping tendency dominate ever attracting tendencies

Classification of solids:

Solids can be classified based on their molecular arrangements and constituent particles in two broader categories,

(i) based on constituent particle & their inter particle forces:
(ii) based on arrangement of particles

[I] based on constituent particle & their inter particle forces:

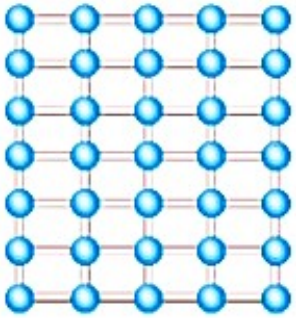
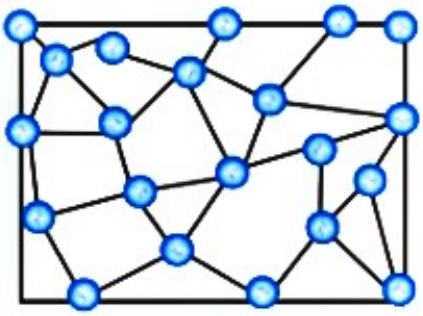
	Solid	Particles	Inter particle forces	Examples	Nature
1.	Molecular solids	Atoms, Non-polar molecules, Polar molecules	London dispersion force, Dipole–dipole force, H (Bond)	eg. crystals of inert gases Ar(S), Kr(S), I_2(S), CH_4(S), H_2O(S)	Soft to hard, Insulators, Low melting and boiling points.
2.	Metallic solids	Metal atoms and electrons	Strong metallic bord	eg. All metal & Alloys (Na, Fe, Cu, Ag etc)	Soft to hard, conductors of heat and electricity, high melting & boiling point
3.	Covalent network solid	atoms	strong covalent bond	e.g. graphite, silica (SiO_2) SiC etc.	Hard, insulators, high melting & boiling point (graphite is good and U)
4.	Ionic solids	Oppositely charged ions	Strong coulombic attraction force	e.g. Nacl, KBr, KNO_3, ZnS etc	Hard, insulators in solid state, High melting & boiling point

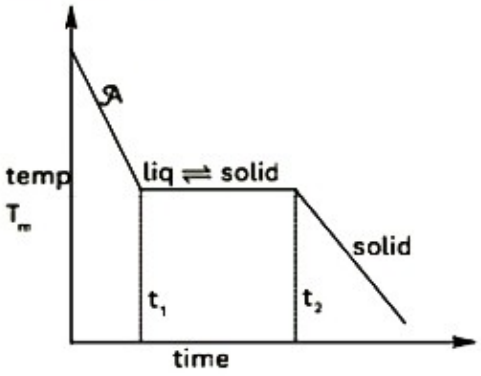
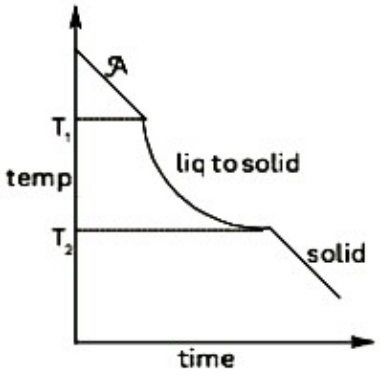
[II] On the basis of arrangement of particles:

Based on arrangement of particles in the lattice, solids can be classified into two categories Crystalline solids and amorphous so

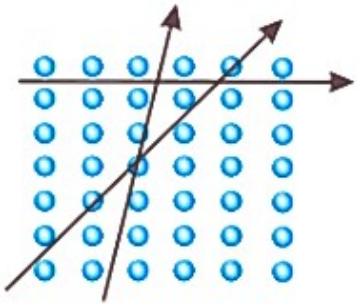
Points to remember

- Graphite, even though it is a covalent solid, is a good conductor of electricity due to presence of unpaired electrons in the lattice.
- Graphite is soft due to its layered structure.
- Ionic solids behave as insulators in solid state while conductors in molten and aqueous solution.

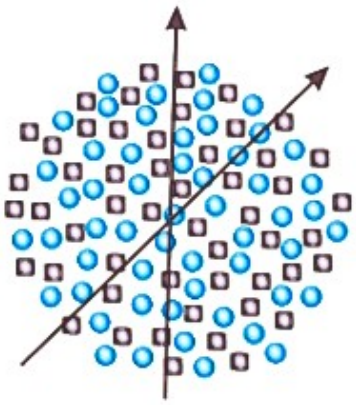
	Crystalline solids		Amorphous solids
1.	In these solids particles are arranged systematically throughout the crystal, known as long range order.  e.g. NaCl, ZnS diamond Fe, Cu etc	1.	In these solids particles are arrange randomly to form short range order.  e.g. glass rubber plastic, amorphous silica.
2.	Crystalline solids are also called True solids	2.	Also called as Pseudo solid or super cooled liquids.
3.	These solids are formed by slow cooling of liquid under controlled conditions so that the particles get opportunity to minimise potential energy & form long range order.	3.	These solids are formed by rapid cooling of liquid so that all particles do not get opportunity to minimise their potential energy due to this short range order is formed.

4.	Sharp melting point which can be understood by below cooling curve.	4.	Range of melting point which can be understood by below cooling curve.
5.	Cooling curve  Sharp melting point T_m	5.	Cooling curve.  Range of melting point T_1 to T_2
6.	Structure of crystalline compounds, sometimes, depend on condition of formation of crystal. Some crystalline compounds may have more than one type of crystalline structure. These structures are called isomorphous structures & the phenomena is known as polymorphism. e.g. $CaCO_3$ → Arogonite, Calcite e.g. ZnS → Zinc blande, Wcrtzite	6.	On slow heating & cooling amorphous solids can be converted to crystalline. For instance, Silica (amorphous) is converted into crystalline silica.

7. **Anisotropy:** In crystalline solids, some physical properties like refractive index, mechanical strength, electrical & thermal conductivity are different in different directions. This property of solid compound is known as Anisotropy and the crystal showing anisotropy is known as anisotropic.



7. **Isotropy:** Due to randomisation physical properties of amorphous solids are same in all direction.



Q1 **Which of the following is correct for ionic crystals?**

(A) They possess low melting point.

(B) They all are non-electrolyte.

(C) The vapour pressures of ionic crystals at ordinary temperature are very low.

(D) They exhibit directional properties of the bond.

Sol. (C)

Ionic crystals have no directional bonds as the oppositely charged ions are attracted by electrostatic forces and arranged in a definite ratio. They have high melting point and all behave as electrolytes. The vapour pressures of ionic crystals at ordinary temperature are very low.

Q2 **Which of the following are the properties of crystalline solid?**

(A) Isotropic **(B) Sharp melting point**

(C) A definite and regular geometry **(D) High intermolecular forces**

Sol. (B,C,D)

Crystalline solids are anisotropic as they have different value of same property in different directions

Q3 **Which of the following are amorphous solids?**

(A) NaCl **(B) CaF_2** **(C) Glass** **(D) Plastics**

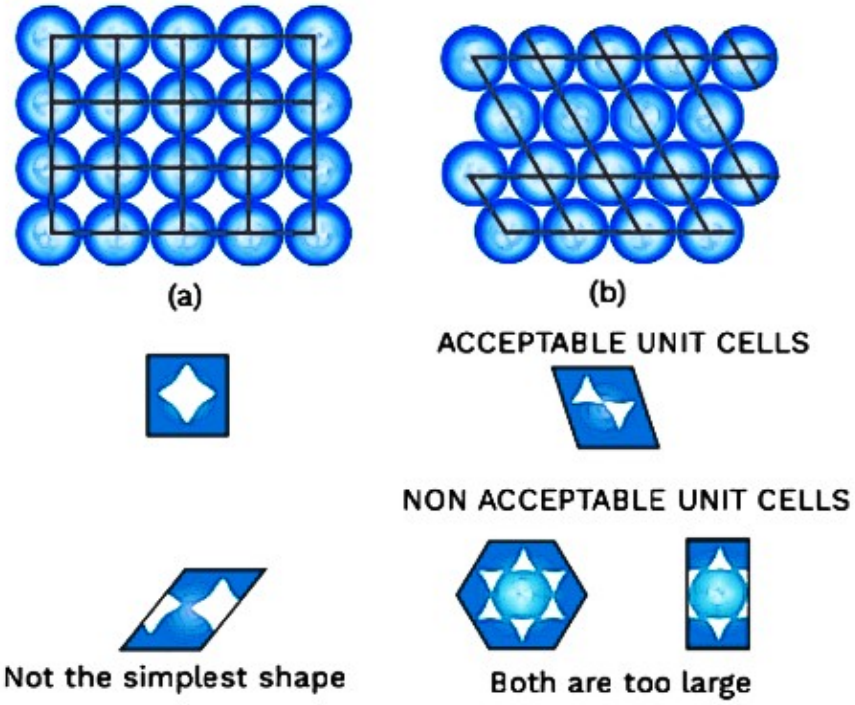
Sol. (C,D)

Both have asymmetric arrangement of atoms.

Basic Definitions in solid state:

1. **Unit cell**

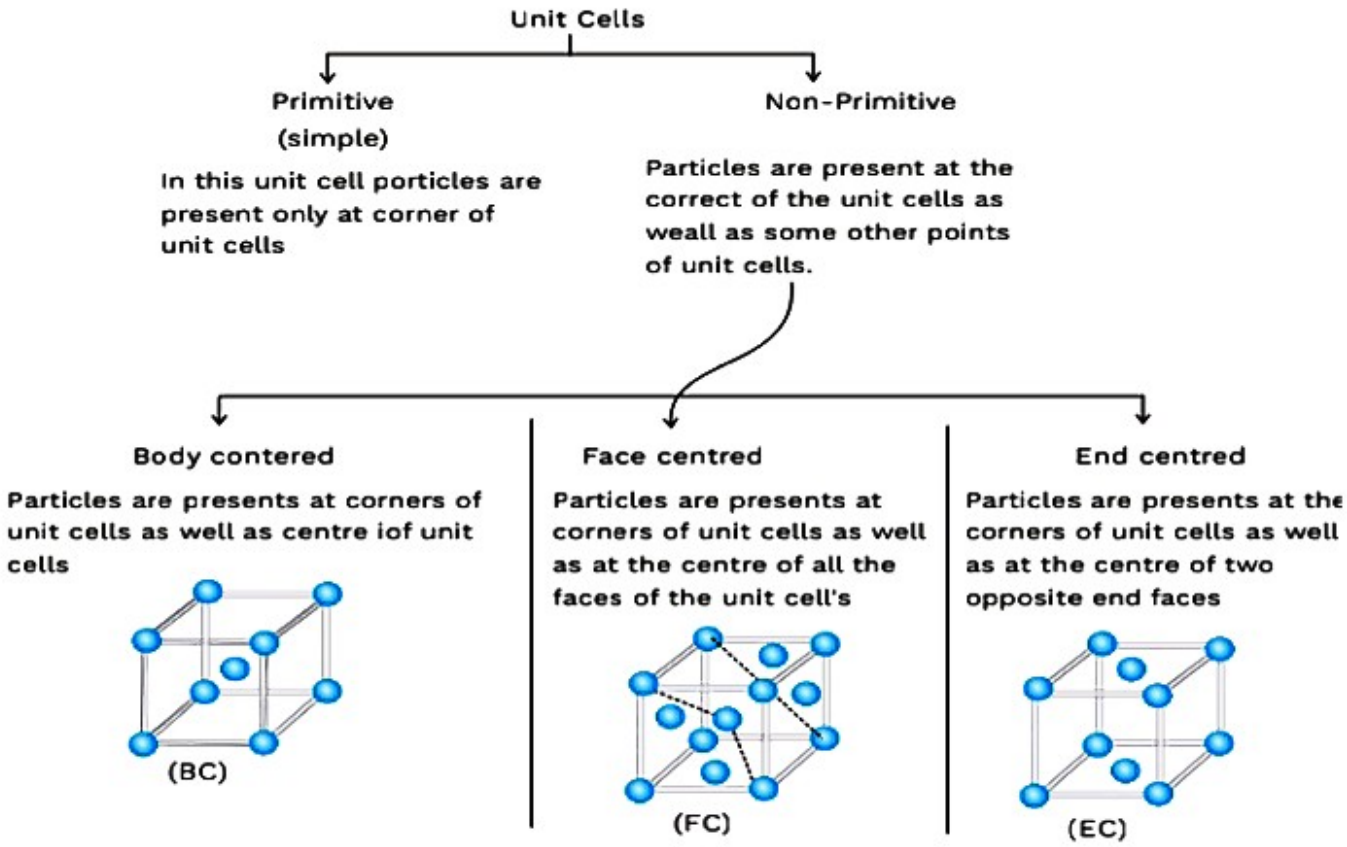
It is the smallest arrangement of particle in the space, which on the repetition forms the whole crystal. A unit cell is the building blocks of a crystal.



A unit cells contain minimum information about crystals in terms of angle & distance.

In a particular crystal system, more than one type of unit cells can be assumed, and the assumptions are made to minimise the calculations.

Classification of unit cells



Unit cells along a line (1D) :

Only one parameter (interparticle distance) is required to specify the unit cell.



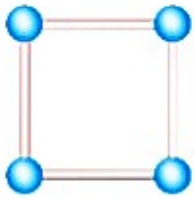
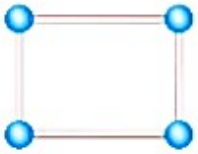
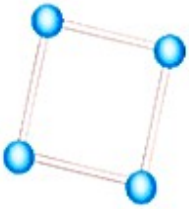
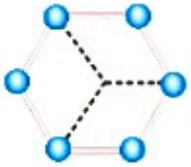
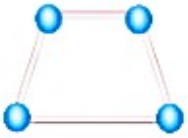
Unit cells in a layer (2D)

Only three parameters are required to specify the unit cell.

Axial parameter i.e., inter particle distances along two axes (a & b)

Angular Parameter i.e., angle between line joining the particles (α)

Classifications of unit cells in 2D

Square	Rectangular	Rhombic	Hexagonal	Parallelo-gram
a = b ∝= 90°	a ≠ b ∝= 90°	a = b ∝ ≠ 90° ≠ 60°	a = b ∝ = 60°	a ≠ b ∝ ≠ 90°
				

2. Space lattice

It is the arrangement of particles in the space which is required to characterise the crystal.

Based on geometry, seven types of crystal system are present. In each crystal system one or more than one type of unit cells can be possible. Unit cells in 3D crystal system are known as **Bravais lattice**, named after French physicist Auguste Bravais.

Total 7 crystal system contain 14 Bravais lattices.

Total six parameters are required to specify unit cells in 3D.

Axial parameters: inter particle distances along three axes (a, b, c)

Angular parameters: angle between line joining the particles in 3D (α, β, γ)

Q4 **Which of the following statements is correct?**
(A) The number of Bravais lattices in which a crystal can be categorised is 12.
(B) There are nine types of primitive unit cells.
(C) Orthorhombic crystal system has five Bravais lattices.
(D)Tetragonal crystal system has two Bravais lattices.

Sol. **(D)**
Tetragonal crystal system has two Bravais lattices - simple and body centered.

Q5 **Which of the following systems have lattice parameters as $a \neq b \neq c$?**
(A) Orthorhombic **(B) Monoclinic** **(C) Triclinic** **(D) Cubic**

Sol. **(A,B,C)**

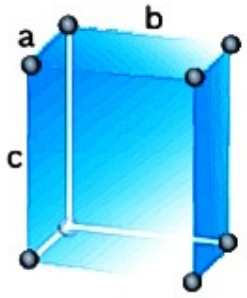
Bravais Lattices

Crystal system	Axial parameters	Angular parameters	Type of Bravais lappics	Example
1. Cubic	A = b = c	$\alpha = \beta = \gamma = 90°$	S, BC, FC	Cu, NaCl diamond etc.
2. Tetragon	a = b ≠ c	$\alpha = \beta = \gamma = 90°$	S, BC	White SnO_2, TiO_2 etc.
3.Orthorhombic	a ≠ b ≠ c	$\alpha = \beta = \gamma = 90°$	S, BC, FC, EC	Rhombic sulphur $CaCO_3$ (aragonite)
4.Rhambohedral	a = b = c	$\alpha = \beta = \gamma \neq 90° < 120°$	S	$CaCO_3$(Calcite) HgS (Cinnabar)
5. Hexagonal	a = b ≠ c	$\alpha = \beta = 90° = \gamma = 120°$	S	Graphite Zn, ZnO, etc.

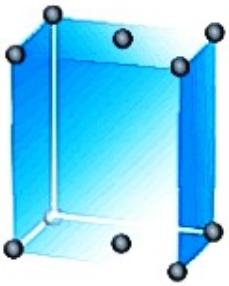
6. Monotonic	$a \neq b \neq c$	$\alpha = \gamma = 90°$ Bt $90° > 90°$	S, BC	Monoclinic sulphur
7.Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq S$	S	$CuSO_2$, $5H_2O$, $K_2Cr_2O_7$

S: Simple unit cell
BC: Body Centred unit cell
FC: Face Centred unit cell
EC: End Centred unit cell

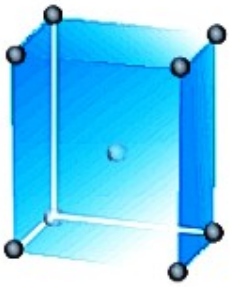
Total 14 lattices



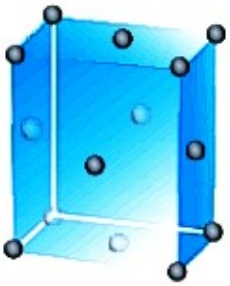
Orthorhombic S



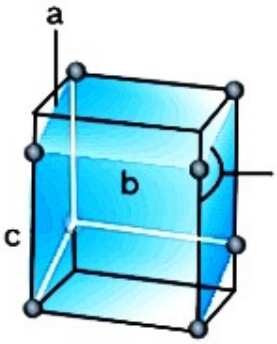
Orthorhomibic EC



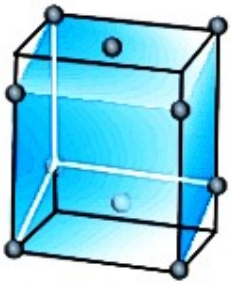
Orthorhomibic BC



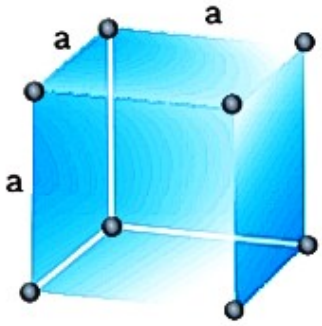
Orthorhomibic FC



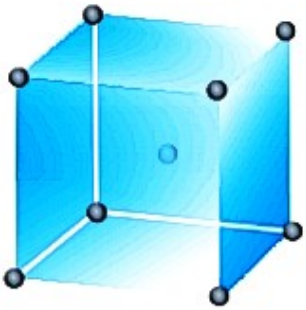
Monoclinic S



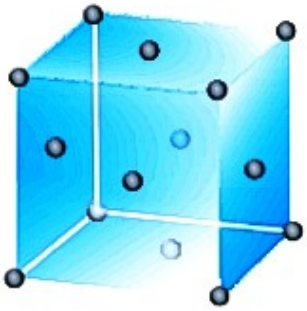
Monoclinic EC



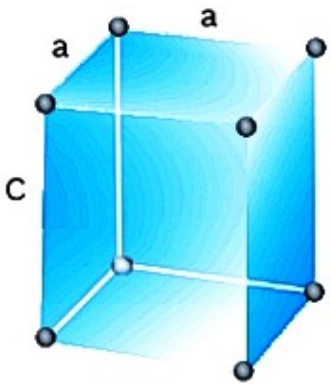
Cubic S



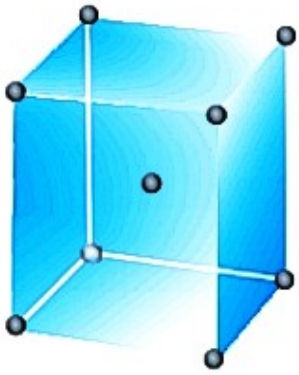
Cubic BC



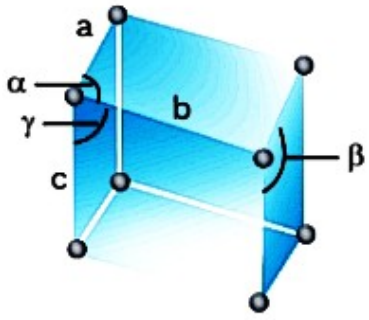
Cubic FC



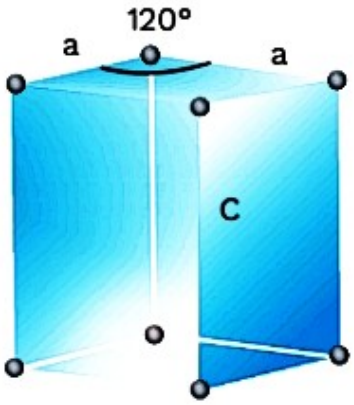
Tetragonal S



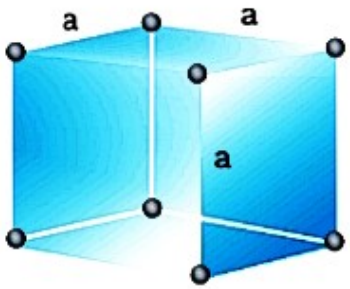
Tetragonal BC



Triclinic



Hexagonal



Trigonal R

Q6 **Which primitive unit cell has unequal edge lengths ($a \neq b \neq c$) and all axial angles different from 90°?**
(A) Monoclinic (B) Triclinic (C) Tetragonal (D) Hexagonal

Sol. **(B)**
Triclinic : Edge lengths ($a \neq b \neq c$), axial angles ($\alpha \neq \beta \neq \gamma \neq 90°$)

Q7 **The unit cell with crystallographic dimensions, $a \neq b \neq c$, $\alpha = \gamma = 90°$, $\beta \neq 90°$ is**
(A) monoclinic (B) tetragonal (C) triclinic (D) orthorhombic

Sol. **(A)**
For monoclinic crystals, $\alpha \neq b \neq c$, $\alpha = \gamma = 90°$, $\beta \neq 90°$

Q8 **Which type of crystals contain the maximum number of Bravais lattices?**
(A) Cubic (B) Triclinic (C)Orthorhombic (D) Tetragonal

Sol. **(C)**
Orthorhombic crystals contain the maximum number of Bravais lattices which includes primitive, fcc, bcc and base centered.

3. Volume of unit cell:

The formula for calculating volume of any type of unit cell is given as

$$V = abc\sqrt{1 - \cos^2\alpha - \cos^2\beta = \cos^2\gamma + 2\cos\alpha\cos\beta\cos\gamma}$$

For cube:

$\alpha = \beta = \gamma = 90°$

$a = b = c$

Points to remember

Cubic crystal system is the most symmetrical crystal system.
Triclinic is the most unsymmetrical crystal system.

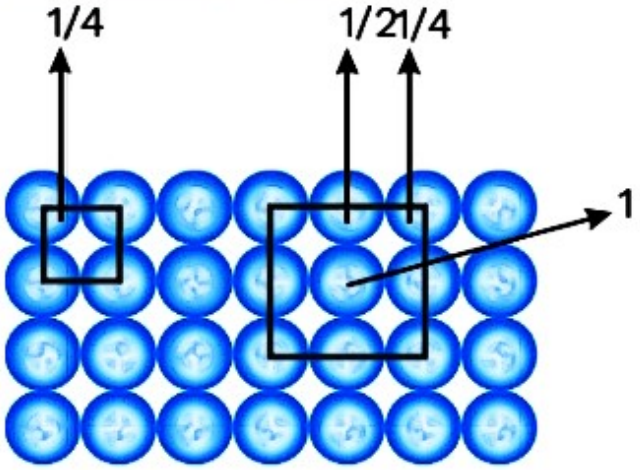
$1cm = 10^8 A°$

The volume generally calculated in cubic centimetre or cubic angstrom.

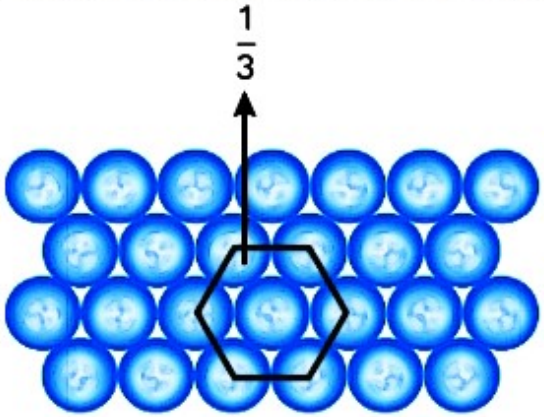
4. Contribution of a particle in a unit cell

Effective number of particles in a unit cell can be determined by slicing the unit cell out of the lattice structure and calculating the portion of particle left inside the unit cell. The diagrams given below will help us understand the ef fective contribution of particles.

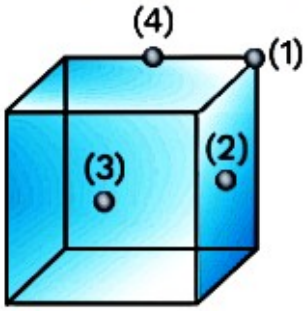
Calculation of 2D lattices



From above figures it is evident that, particles at the corners of the square will have 1/4th part of it inside the square. While particles which lie inside the unit cell will have 1 contribution per unit cell. Similarly, in the diagram given below contribution of particles at each corner of hexagon will be 1/3 per unit cell.



Calculations for 3D lattices in a cubic crystal:



(1) Contribution of particle, present at any corner of a cubic unit cell, in one unit cell $= \frac{1}{8}$

(2) Contribution of particle, present at any face centre of a cubic unit cell, in one unit cell $= \frac{1}{2}$

(3) Contribution of particle, present at body centre of a unit cell, in one unit cell = 1

(4) Contribution of particle, present at edge centre of a unit cell, in one unit cell $= \frac{1}{4}$

Q9 **An ionic compound has a unit cell consisting of A ions at the corners of a cube and B ions on the centres of the faces of the cube. The empirical formula for this compound would be**

(A) AB **(B) A_2B** **(C) AB_3** **(D) A_3B**

Sol.

(C)

Number of A ions per unit cell $= \frac{1}{8} \times 8 = 1$

Number of B ions per unit cell $= \frac{1}{2} \times 6 = 3$

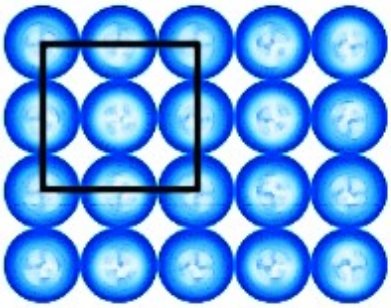
Empirical formula = AB_3

5. Unit cell content (Z):

The effective no. of particles in one unit cell is known as unit cell content (denoted by Z)

Mass of one unit cell = (molecular mass × Z) amu

For instance, in a **2D lattice given below**



Unit cell content will be

= 1 (For particle present at centre) $+\left(\frac{1}{2}\times 4\right)$ (for particles present at edges of unit cell) $+\left(\frac{1}{4}\times 4\right)$ (for particles present at corners) = 4

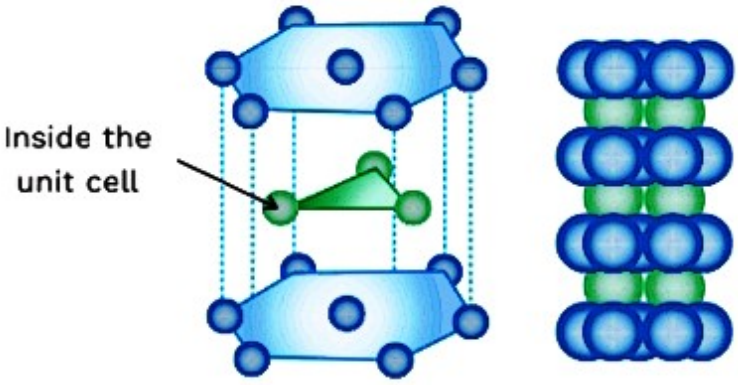
In cubic unit cell:

Unit cell content of simple cubic (SC) $= \frac{1}{8}\times 8 = 1$

Unit cell content of body centred (BCC) = 1 + 1 = 2

Unit cell content of face centred cubic (FCC) $= 1+\left(\frac{1}{2}\times 6\right) = 4$

In a 3D hexagonal unit cell



Unit cell content $=3+\left(\frac{1}{2}\times 2\right)+\left(12\times\frac{1}{6}\right)=6$

mass of one SC unit cell = M amu

mass of one BCC unit cell = 2 M amu

mass of one FCC unit cell

Where M is mass of constituent particle in amu and grams for given equations respectively.

Q10 **Na and Mg crystallize in bcc and fcc type crystals respectively, then the number of atoms of Na and Mg present in the unit cell of their respective crystal is**

(A) 4 and 2 **(B) 9 and 14** **(C) 14 and 9** **(D) 2 and 4**

Sol. **(D)**

In bcc atoms are present at corners and one in the centre of the unit cell.

$\therefore$ Number of atoms per unit cell $=8\times\frac{1}{8}+1=2$

In fcc atoms are present at the corners and also centre of the six faces of each cell.

$\therefore$ Number of atoms per unit cell $=8\times\frac{1}{8}+6\times\frac{1}{2}=4$

11. **In a face centred cubic arrangement of A and B atoms, A atoms are at the corners of the unit cell and B atoms at the face centres. One of the A atom is missing from one corner in the unit cell. The simplest formula of the compound is**

(A) A_7B_8 **(B) AB_3** **(C) A_7B_{24}** **(D) A_7B_3**

Sol. **(C)**

If one atom (A) is missing from one corner,

number of atoms A in unit cell $= 7 \times \frac{1}{8} = \frac{7}{8}$

Number of atoms B in unit cell $= 6 \times \frac{1}{2} = 3$

$A : B = \frac{7}{8} : 3$ so simplest formula is A_7B_{24}.

12. **If three elements X, Y and Z crystallized in cubic solid lattice with X atoms at corners, Y atoms a each face centre and Z-atoms at the edges, then the formula of the compound is**

(A) XYZ **(B) XY_3Z** **(C) XY_3Z_3** **(D) X_3YZ**

Sol. **(C)**

Atom X is shared by 8 corners. 1 atom of Z is shared by 4 unit cells. Atom Y is present at centre of the face and 1 atom of Y is shared by 2 unit cell.

Hence, effective number of atoms of X per unit cell $= 8 \times \frac{1}{8} = 1$ Effective number of atoms of Z per unit cell $= \frac{12}{4} = 3$

Effective number of atoms of Y per unit cell $= \frac{1}{2} \times 6 = 3$ Hence, the formula of the compound is XY_3Z_3.

6. Density of crystal:

Density of crystal = density of unit cell $= \frac{\text{mass of unit cell}}{\text{vol. of unit cell}}$

$$d = \left(\frac{M \times Z}{V}\right) \frac{amu}{\mathring{A}^3}$$

$$d = \left(\frac{M \times Z}{N_A \times V}\right) gm/cm^3$$

Q13. **An element crystallizes in FCC (molar mas = 60) Edge length of its unit cell = 10 A° Cal density of solid in (1) am/Å³ ; (2) gm/cm²**

Sol: (1) $d=\frac{4\times 60}{(10)^3}=0.24\,\frac{amu}{A^\circ}$

(2) $d=\frac{4\times 60}{6\times 10^{23}\times\left(10\times 10^{-8}\right)^3}=0.4gm/cm^3$

Q14. **At 100°C, copper (Cu) has *fcc* unit cell structure with cell edge length of *x* Å. What is the approximate density of Cu (in g cm⁻³) at this temperature? [Atomic mass of Cu = 63.55 u]**

(a) $\frac{205}{\chi^3}$ **(b)** $\frac{211}{\chi^3}$ **(c)** $\frac{105}{\chi^3}$ **(d)** $\frac{422}{\chi^3}$

Sol. **(D)**

$d=\frac{ZM}{N_A a^3}$ [For fcc, Z = 4]

$=\frac{4\times 63.55}{6.023\times 10^{23}\times x^3\times 10^{-24}}=\frac{42.20}{x^3}\times 10=\frac{422}{x^3}$

7. Packing fraction & void fraction:

Fraction of unit cell occupied by particles in each unit cell is known as **packing fraction (P.F.)**

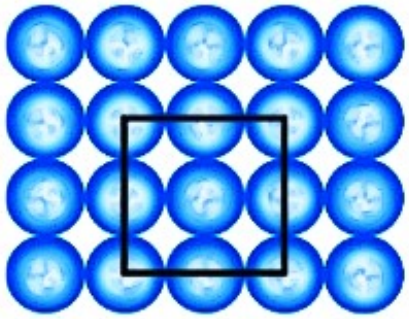
Void fraction is fraction of empty spaces in given unit cell.

Void fraction = 1- packing fraction.

In 2-D unit cell $\boxed{P.F.=\frac{\text{area of particles inside unit cell}}{\text{total area of unit cell}}}$

In 3-D unit cell $\boxed{P.F.=\frac{\text{volume of particles inside unit cell}}{\text{total volume of unit cell}}}$

For instance, in a 2-D unit cell given below:



Radius of circle = r

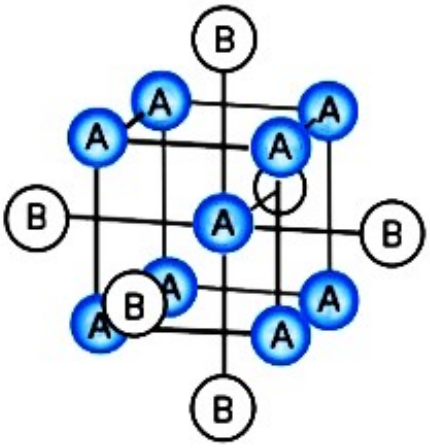
$$= \frac{16r^2}{\pi r^2 + 2\pi r^2 + \pi r^2}$$

$$= \frac{16r^2}{4\pi r^2} = \frac{4}{\pi}$$

$$\text{P.f.} = \frac{\pi}{4}$$

(8) Co- ordination number (CN) Or Nearest neighbours

It is no of nearest particles in contact with any particle. Particles which are at higher distances are called next nearest neighbours, next to next nearest neighbours and so on.



For instance, in the above diagram coordination number of A with respect to B will be 6 as six particles of B surround A.

(9) Formula of solid

Empirical formula of solid is the representation of all the particles in a unit cell with proper no.

Let's understand this by given examples

Ex 1:

In a cubic unit cell of X and Y atoms

X atom are at the corners

Y atom are at the body centres.

$\therefore$ Formula of solid = XY

Ex 2:

A solid consist of X & Y atoms

X atoms are at the corners

Y atoms are at face centres

$\therefore$ Formula of solid = XY_3

Q15. Ca (MW= 40) forms FCC unit cell. A cubic crystal of Ca having mass = 4g and edge length of unit cell = 10 A°
Calculate:
(i) density of solid
(ii) No. of unit cells in the crystals
(iii) Dimension of cubic crystal (edge of cube) $\left(N_A = 6 \times 10^{23}\right)$

Sol: (1) $d = \dfrac{4 \times 40}{N_A \times \left(10 \times 10^{-8}\right)^3}$

$= \dfrac{4 \times 40}{6 \times 10^{23} \times 1000 \times 10^{-24}}$

$= \dfrac{4 \times 40}{6 \times 100} = \dfrac{16}{60}$ gm/cm

(2) No. of moles $= \dfrac{4}{40} = 0.1$

No. of Ca atom $= 0.1\ N_A = x$

No. of FCC unit cells $= \dfrac{0.1N_A}{4} = \dfrac{x}{4} = 1.5 \times 10^{22}$

(3) $\left(10 \times 10^{-8}\right)^3 \times 1.5 \times 10^{22} = a^2$

Q16. **The perovskite structure is a cubic lattice, with Ba^{2+} ions occupying the corners of the unit cell, O^{2-} ions occupying the face centred and Ti^{4+} ions occupying the centres of the unit cell. Thus the formula is $Ba_x Ti_y O_z$. The sum of x, y and z is**

Sol. **(5)**

Number of Ba^{2+} ions $\frac{1}{8} \times 8$ corners = 1 Number of Ti^{4+} ions = 1 body centre atom

Number of O^{2-} ions = $6 \times \frac{1}{2} = 3$

Therefore, the formula is $Ba^{2+}\ Ti^{4+}\ O_3^{2-}$ □ $BaTiO_3$

Hence, x + y + z = 1 + 1 + 3 = 5

Q17. **Chromium metal crystallizes with a body centred cubic lattice. The length of the unit cell edge is found to be 287 pm. What would be the density of chromium in g/cm³?**

Sol. **(7.30)**

$$\text{Density, } \rho = \frac{n \times \text{at.wt.}}{V \times N_0} = \frac{n \times \text{at.wt}}{N_0 \times a^3}$$

For *bcc*, $n = 2$, given: $a = 287 \times 10^{-10}$ cm

□ Density = $\frac{2 \times 51.99}{(287 \times 10^{-10}\ \text{cm})^3 \times 6.023 \times 10^{23}} = 7.30$ g/cm³

18. **A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 400 pm. If the density of the substance in the crystal is 8 g cm⁻³, then the number of atoms present in 256 g of the crystal is $N \times 10^{24}$. The value f N is_____.**

Sol. **(2)**

$$d = \frac{Z \times M}{a^3 \times N_A}$$

For *fcc* lattice,
Number of atoms per unit cell, $Z = 4$

Substituting values in equation, we get

$$8 = \frac{4 \times M}{(400 \times 10^{-10})^3 \times 6.023 \times 10^{23}}$$

$$M = \frac{8 \times (400 \times 10^{-10})^3 \times 6.023 \times 10^{23}}{4}$$

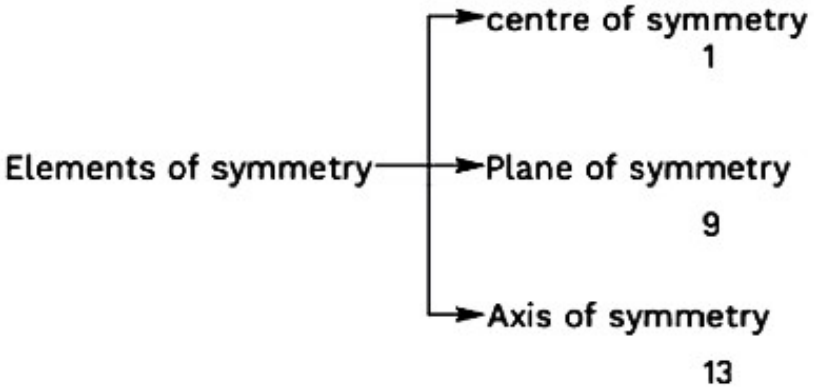
$\therefore$ M = 77.0944 g mol^{-1}

77.0944 g of solid has 6.023×10^{23} atoms

$$= \frac{6.023 \times 10^{23}}{77.0944} \times 256 \text{ atoms} = 20 \times 10^{23} = 2 \times 10^{24}$$

Comparing it with $N \times 10^{24}$, we get N = 2

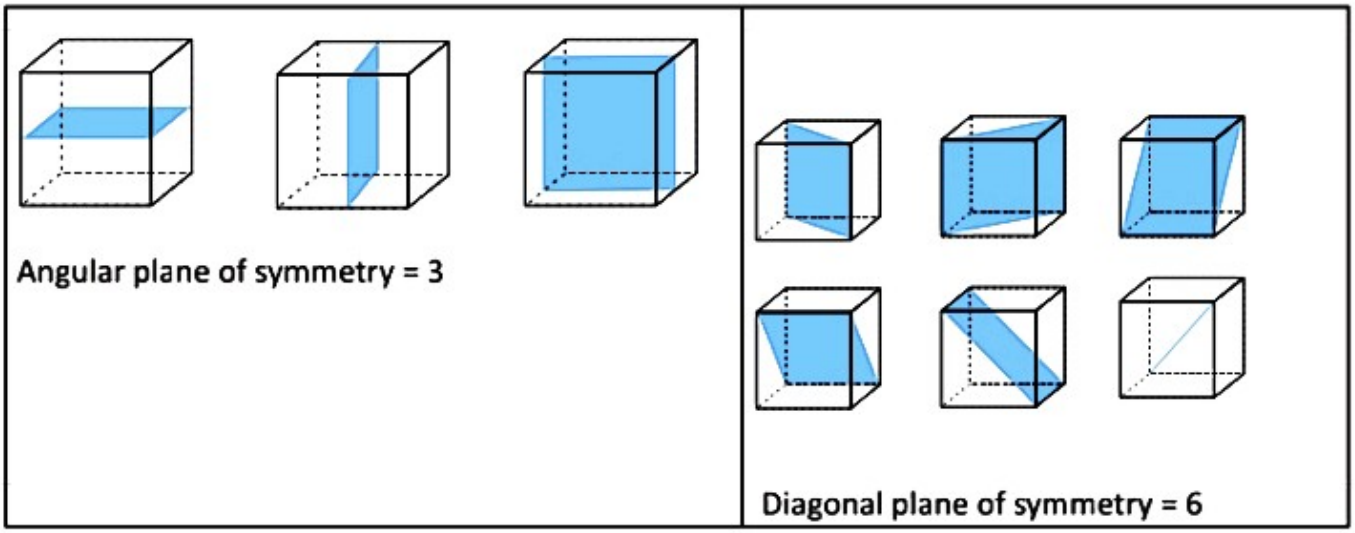
Elements of symmetry in cubic crystal



Total elements of symmetry in cubic crystal = 1 + 9 + 13 = 23

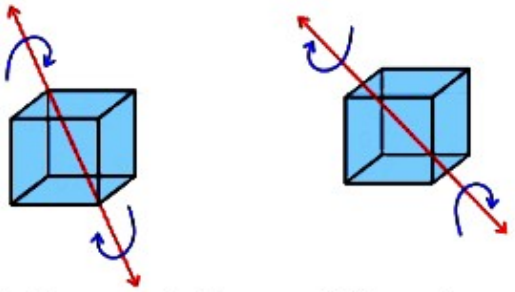
Plane of symmetry

A plane, real or imaginary, passing through an object in such a way that it divides the object in two equal halves is called plane of symmetry.



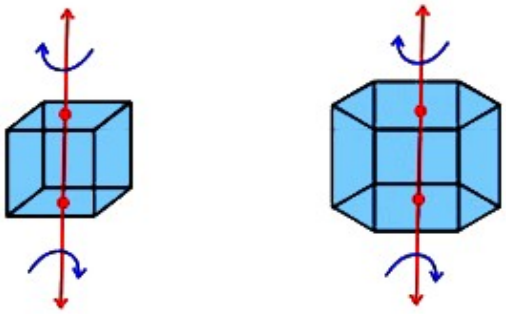
Axis of symmetry

An axis, real or imaginary, along which if we rotate any object with a particular angle (θ) and we get a geometry similar to the original object is termed as axis of symmetry.
If we define a number n, where n = , then a symmetry achieved at angle θ is known as n-fold (C_n) axis of symmetry.



(i) Six two fold symmetries

(ii)four three fold symmetries



(iii) Three four fold symmetries

(iv)One six fold symmetries

FORMATION OF METALLIC CRYSTALS:

Assumption made for formation of metallic crystals are as follow

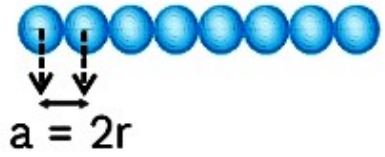
All the atoms are identical spheres of radius 'r'

Atoms are arranged to minimise their potential energy. So, they are arranged as close to each other as possible.

During the formation of crystal, first edge is formed followed by formation of plane and whole crystal.

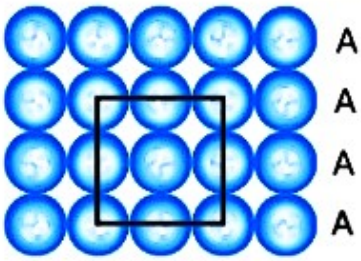
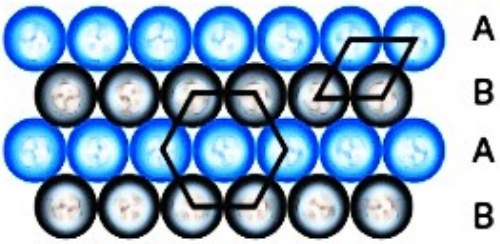
Formation of edge (1 D)

Particles are arranged along a line to form edge of a lattice as shown below.



Formation of plane (2D)

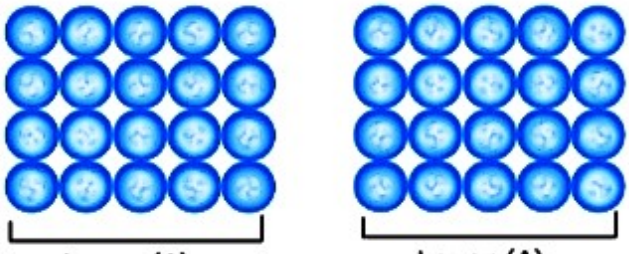
A plane can be formed in two different ways namely square and hexagonal as discussed below.

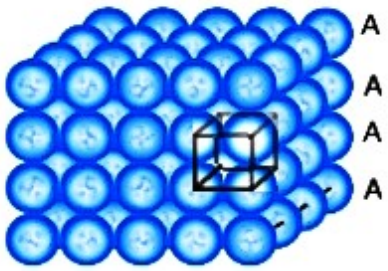
Square (2D) packing	Hexagonal (2D) packing
	
Coordination number = 4	Coordination number = 6
Unit cell content = 1	Unit cell content = 3
Packing fraction	Packing fraction
$= \frac{\pi r^2}{4r^2}$	$= \frac{\left(2 \times \pi r^2\right) + \pi r^2}{6 \times \left(\frac{\sqrt{3}}{4} \times a^2\right)}$
$= \frac{\pi}{4}$	$= \frac{\pi}{2\sqrt{3}}$
$= 0.78$	$= 0.91$
78% area is occupied & 22% is vacant	91% area is occupied 9% is vacant

From above discussion it is cleared that hexagonal (2D) packing is more efficient than square (2D) packing.

Formation of 3-D packing form 2-D square sheet

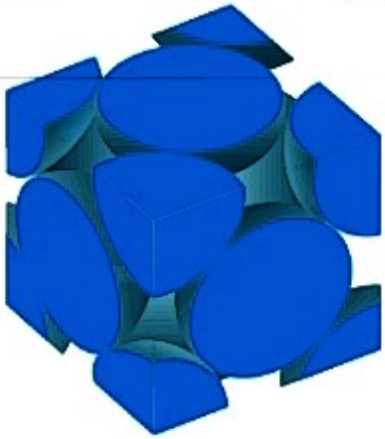
(1) AA type packing (Simple cubic (SC))





$\frac{1}{8}$ atom at
8 corners

$\frac{1}{2}$ atom at
6 Faces



Face-centered cubic metal
4 atoms per unit cell

Cell parameters:
Edge length = a = 2r

Unit cell content of simple carbic (Z) = $\frac{1}{8} \times 8 = 1$

Mass of one unit cell = M amu

$$= \frac{M}{N_A} g$$

Volume of unit cell $V = a^3 = 8r^3$

$$\text{density of simple cubic} = \frac{1 \times M}{V} \frac{\text{amu}}{\text{Å}^3}$$

$$= \frac{1 \times M}{N_A \times V} \text{gm/cm}^3$$

P.F. = volume occupied by spheres in the unit cell / volume of unit cell.

$$= \frac{1 \times \frac{4}{3}\pi r^3}{a^3} = \frac{4\pi r^3}{3 \times 8r^3}$$

$$= \frac{\pi}{6}$$

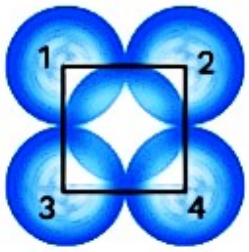
$$\approx 0.52$$

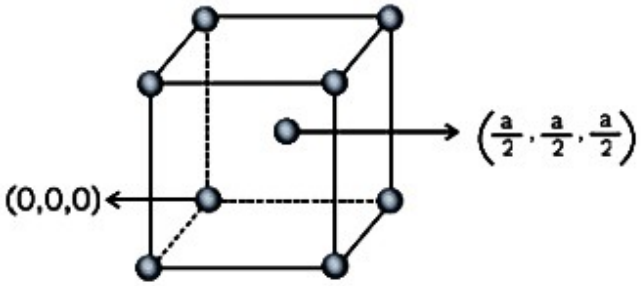
Void fraction = 0.48

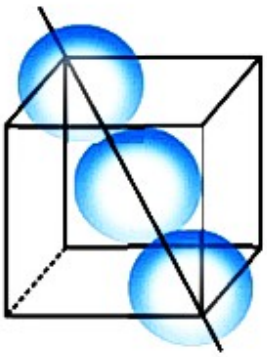
52% of unit cell is occupied by sphere & 48% is vacant.

Neighbours	Distance	Number
Nearest	$2r = a$	8
Next nearest	$a\sqrt{2}$	12
Next to next nearest	$a\sqrt{3}$	$\left(\frac{1 \times 8}{1}\right) = 8$

Special case of squared sheet: body centred cubic (BCC)

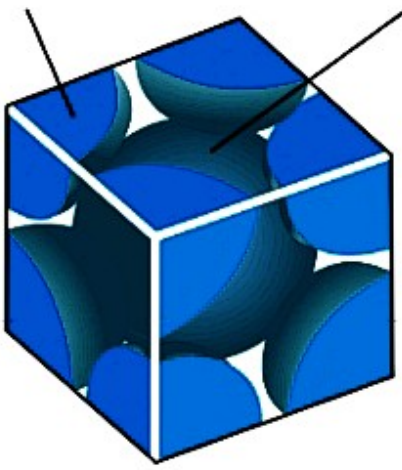






$\frac{1}{8}$ atom at
8 corners

1 atom at
Body Center



Cell parameter: $\sqrt{3}a = 4r$

Unit cell content (z): $\left(\frac{1}{8}\times 8\right) + 1 = 2$

Mass of one unit cell: $2M\,amu = \frac{2\times M}{N_A}\,gm$

Volume of one unit cell: $V = a^3 = \left(\frac{4r}{\sqrt{3}}\right)^3 = \frac{64r^3}{3\sqrt{3}}$

Density of unit cell: $d = \frac{2M}{N}\frac{amu}{\mathring{A}^3}$

$$= \frac{2M}{N_A V} \text{ gm/cm}^3$$

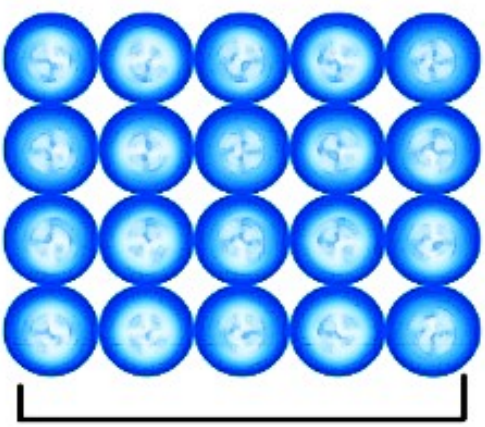
$$\text{P.F.} = \frac{2 \times \frac{4}{3}\pi r^3}{a^3} = \frac{2 \times 4 \times \pi r^3}{3 \times \frac{64 r^3}{3\sqrt{3}}} = \frac{\sqrt{39}}{8} = 0.68$$

68% of unit cell is occupied & 32% is vacant.

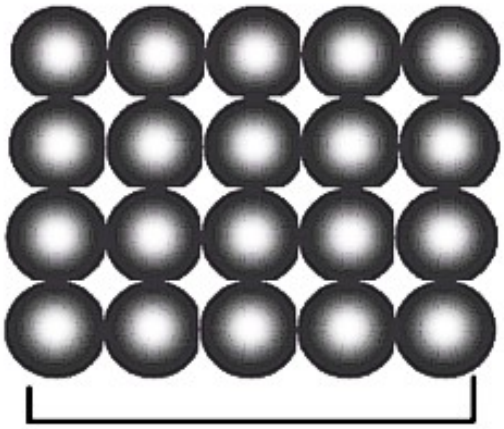
Neighbours	Distance	Number
Nearest (C. No.)	$2r = \frac{\sqrt{3}}{2} a$	$8 = \left(\frac{1 \times 8}{1}\right)$
$(\text{Next})^1$	A	6 (as in SC)
$(\text{Next})^2$	$a\sqrt{2}$	12 (as in SC)
$(\text{Next})^3$	$a\sqrt{\frac{11}{4}}$	24

Closest Packings

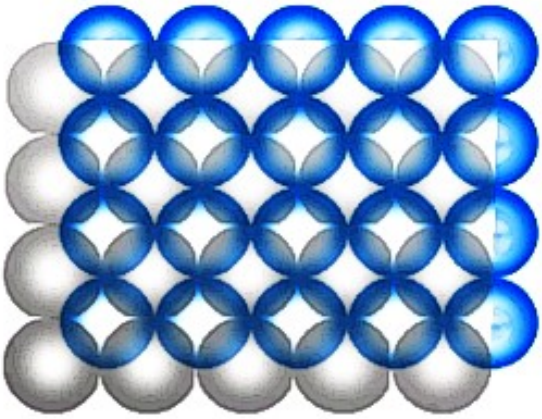
3-D PACKING TO FORM HEXAGONAL CLOSED PACKED STRUCTURE

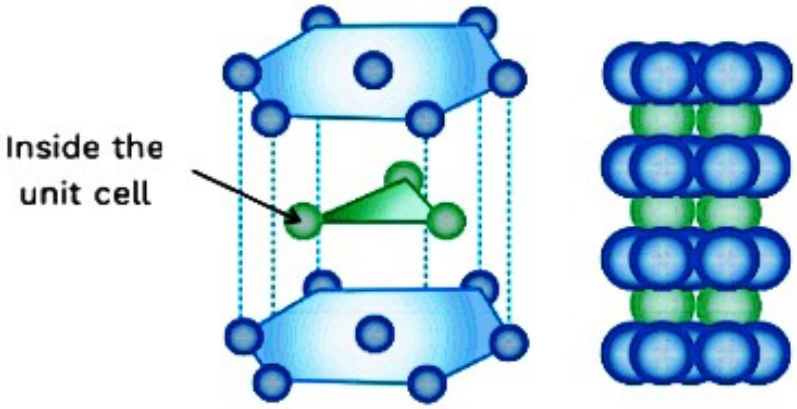


Layer (A)



Layer (A)





The effective contribution of particles in the middle in the unit cell =3

Unit cell content = 6

Contribution from corners = 12 x $\frac{1}{6}$

Contribution from faces = 2 x $\frac{1}{2}$

Contribution from particles in the middle = 3

Mass of one unit cell = 6M amu = $\frac{6 \times M}{N_A}$ gm

Height of one HCP unit cell

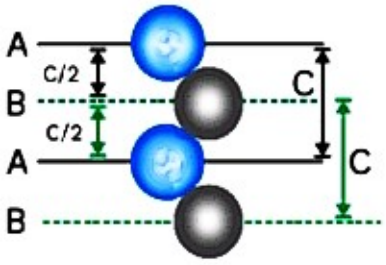
$$C = \sqrt{\frac{2}{3}} \times 4r$$

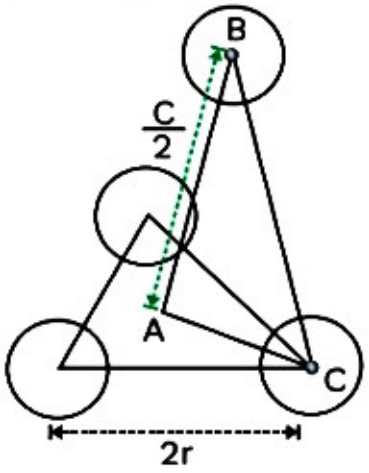
Volume of unit cell = $24\sqrt{2}\, r^3$

Coordination number or nearest neighbours = 12

3 +	6 +	3
above the later	within the layer	below the layer

Point to understand: Calculation of cell parameters of HCP lattice





$$AB = \frac{C}{2}$$

$BC = 2r$

$\angle BAC = 90°$

$$AC = \frac{2r}{\sqrt{3}}$$

In right angled triangle ABC

$$AB^2 + AC^2 = BC^2$$

$$\left(\frac{C}{2}\right)^2 + \left(\frac{2r}{\sqrt{3}}\right)^2 = (2r)^2$$

$$\Rightarrow \frac{C}{2} = 2r\sqrt{\frac{2}{3}}$$

$$\boxed{\therefore C = 4r\sqrt{\frac{2}{3}}}$$

$\therefore$ Distance between 3 consecutive closed packed layer = C
Distance between 2 consecutive closed packed layer = C/2

Q19. **Total volume of atoms present in a face-centred cubic unit cell of a metal is (*r* is atomic radius).**

(a) $\frac{20}{3}\pi r^3$ **(b)** $\frac{24}{3}\pi r^3$ **(c)** $\frac{12}{3}\pi r^3$ **(d)** $\frac{16}{3}\pi r^3$

Sol. **(D)**

In case of a face-centred cubic structure, since four atoms are present in a unit cell, hence volume

$$V = 4\left(\frac{4}{3}\pi r^3\right) = \frac{16}{3}\pi r^3$$

Q20. **Percentages of free space in cubic close packed structure and in body-centred packed structure are respectively**
(A) 48% and 26% **(B) 30% and 26%** **(C) 26% and 32%** **(D) 32% and 48%**

Sol. **(C)**
The packing efficiency in a *ccp* structure = 74%
∴ Percentage free space = 100 − 74 = 26%
Packing efficiency in a body-centred structure = 68% Percentage free space = 100 − 68 = 32%

Q21. **In spinel structure, O^{2-} ions are cubic closed packed, whereas 1/4th of the tetrahedral holes are occupied by A^{2+} cations and half of the octahedral holes are occupied by cations B^{3+}. The empirical formula of this compound is**
(A) A_2BO_4 **(B) ABO_2** **(C) A_2B_4O** **(D) A_4B_2O**

Sol. **(B)**
No. of O^{2-} ions in *ccp* arrangement = 4 No. of octahedral holes = 4

No. of B^{3+} ions = $4 \times \frac{1}{2} = 2$

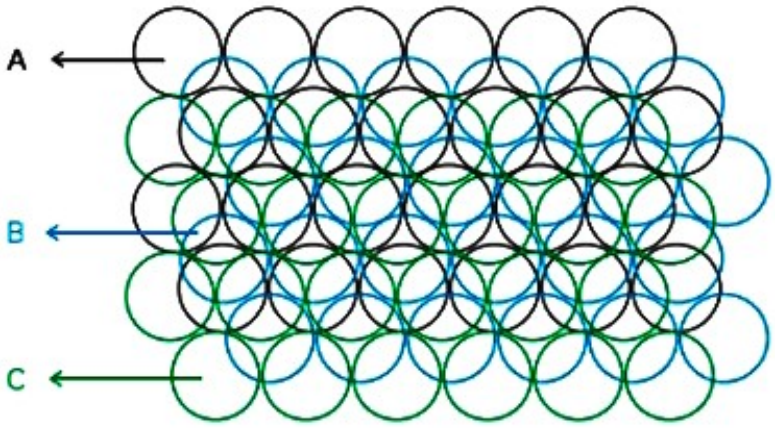
No. of tetrahedral holes = 4 × 2 = 8

No. of A^{2+} ions = $8 \times \frac{1}{4} = 2$

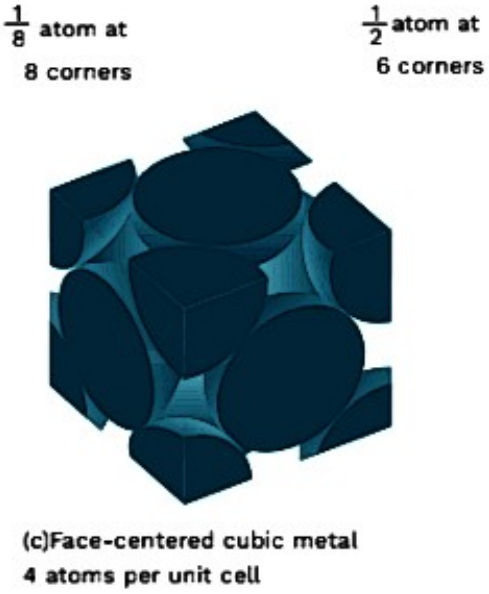
∴ formula = $A_2B_2O_4$ or ABO_2

3D CUBIC CLOSE PACKING (CCP):

ABC..... ABC type packing



Face centred carbic (FCC)



(c)Face-centered cubic metal
4 atoms per unit cell

CCP ≅ FCC

One FCC unit cell contain 4 closest packed layer.

$4r = a\sqrt{2}$

Unit cell content (Z) = 4

Mass of unit cell = 4M amu or $\frac{4M}{N_1}$ g

Vol. of unit cell = $16\sqrt{2}\,r^3$

P.F. = 0.74

V.F = 0.26

Coordination no = 12

Calculation of neighbours

Neighbours	Distance	No.
Nearest (C. No.)	$2r = \frac{a}{\sqrt{3}}$	$12 = \left(\frac{3 \times 8}{2}\right)$
$(Next)^1$	A	6
$(Next)^2$	$a\sqrt{\frac{3}{2}}$	$24 = \left(\frac{3 \times 8}{1}\right)$
$(Next)^3$	$a\sqrt{2}$	12

Points to remember:

(1) FCC & HCC are formed by closest packing of hexagon 2D sheets, so their properties like packing fraction, void fractions, number of nearest neighbours, are same.

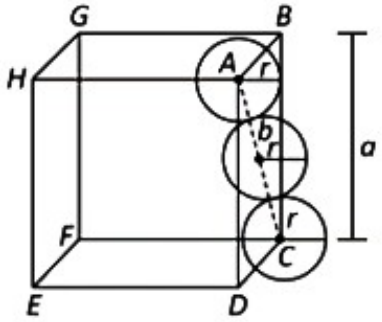
(2) Another generation of FCC

C.NO. of FCC = $\underset{\text{above the layer}}{3} + \underset{\text{within the layer}}{6} + \underset{\text{below the layer}}{3}$

= 12 (according to hexagon 2D sheet)

= $\underset{\text{above the layer}}{4} + \underset{\text{within the same layer}}{4} + \underset{\text{below the layer}}{4}$

= 12 (according to square 2D sheet)



22. **A metal crystallises in a face centred cubic structure. If the edge length of its unit cell is 'a', the closest approach between two atoms in metallic crystal will be**

(A) $\sqrt{2a}$ **(B)** $\frac{a}{\sqrt{2}}$ **(C)** 2a **(d)** $2\sqrt{2a}$

Sol. **(B)**

For *fcc*,
In ΔABC
$AC^2 = b^2 = BC^2 + AB^2 = a^2 + a^2 = 2a^2$
then $b = 4r = \sqrt{2}a$

$$a = \frac{4r}{\sqrt{2}} = 2\sqrt{2}r \Rightarrow r = \frac{a}{2\sqrt{2}}$$

Therefore, distance of closest approach

$$= 2r = 2 \times \frac{a}{2\sqrt{2}} = \frac{a}{\sqrt{2}}$$

Q23. **The *hcp* and *ccp* structure for a given element would be expected to have**
(A) the same coordination number **(B) same pattern of stacking spheres**
(C) the same packing fraction **(D) all the above.**

Sol. **(A,C)**
Both *hcp* and *ccp* structure have same coordinations number and same packing fraction.

Q24. **The correct statements for cubic close packed (*ccp*) three dimensional structure are**
(A) the number of the nearest neighbours of an atom present in the top most layer is 12
(B) the efficiency of atom packing is 74%
(C) the number of octahedral and tetrahedral voids per atom are 1 and 2, respectively
(D) the unit cell edge length is $2\sqrt{2}$ times the radius of the atom.

Sol. **(B,C,D)**
(A) The number of the nearest neighbours of an atom present in the topmost layer is 9 as a sphere is in contact with 6 other spheres in its own layer and it also touches directly 3 spheres in the layer below.
(B) For *ccp*, packing fraction = 74%. In *ccp*, there are two tetrahedral voids per sphere and one octahedral void per sphere. (d) For *ccp*, $a = \frac{4}{\sqrt{2}}r = 2\sqrt{2}r$

Q25. **Which of the following statements are incorrect for a cubic close-packed structure?**

(A) It has ABCABC... type arrangement

(B) There are two type of holes.

(C) It has ABABAB... type arrangement.

(D) *ccp* is also known as *bcc* having packing efficiency of 74%.

Sol. **(C,D)**

There are two types of holes, tetrahedral and octahedral. It has ABCABC... type arrangement. *ccp* is also known as *fcc*.

STUDY OF VOIDS:

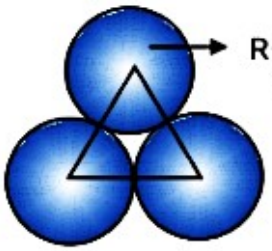
SC	BCC	FCC CCP	HCP
PF → 0.52	0.68	0.74	0.74
VF → 0.48	0.32	0.26	0.26

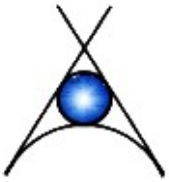
In any unit cell, 100% of space is not occupied by the spheres. The vacant space is known as intestinal space. Different kinds of voids are found in these intestinal spaces.

Voids	Formed at
Triangular	the centre of triangle
Tetrahedral	the centre of tetrahedral
Square	the centre of square

Octahedral	the centre of octahedron
Cubic	the centre of cube

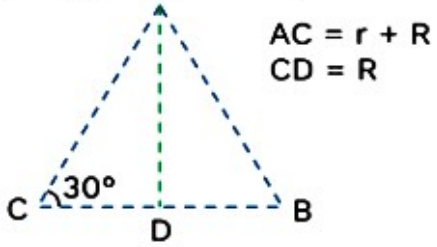
(i) Triangular voids (within hexagon sheet) :





r → radius of largest sphere that can fit into triangular void without destruction.
i.e., radius of void.

$$\frac{r}{R} = \left(\frac{2}{\sqrt{3}} - 1\right) = 0.155$$

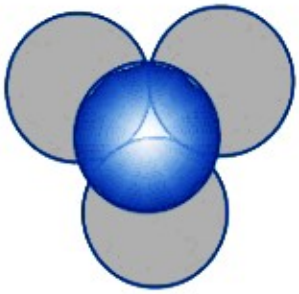


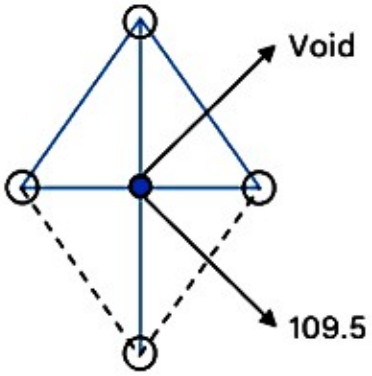
(2) Square voids (2D) (within square sheet) :



$$\boxed{\frac{r}{R} = \left(\sqrt{2} - 1\right) = 0.414}$$

(3) Tetrahedral void:



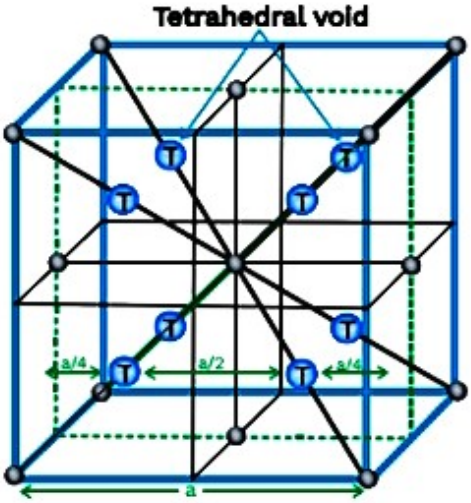


$$\cos\left(109.5\right) = -\frac{1}{3}$$

$$\cos\left(109.5\right) = \frac{\left(r+R\right)^2 + \left(r+R\right)^2 - \left(2r\right)^2}{2\left(r+R\right)^2}$$

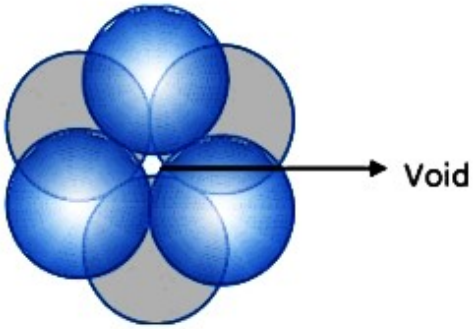
$$-\frac{1}{3} = 1 - \frac{2r^2}{\left(r+R\right)^2}$$

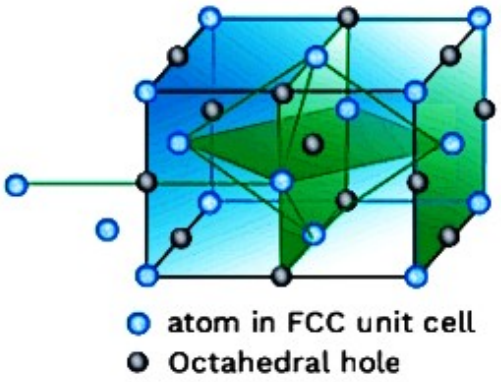
$$\boxed{\frac{r}{R} = \sqrt{\frac{3}{2}} - 1 = 0.225}$$



Tetrahedral Void In One Unit Cell

(4) Octahedral Void

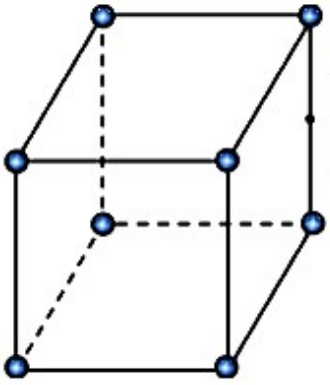




$$\frac{r}{R} = \sqrt{2} - 1 = 0.414$$

(5) Cubic void

At the centre of SC



$$\frac{r}{R} = \sqrt{3} - 1 = 0.752$$

Points to remember

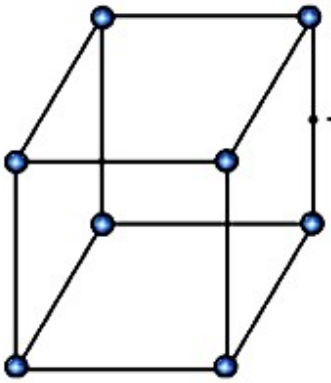
In closest packed arrangement if z is the no. of particles per unit cell, then Total number of tetrahedral voids in such arrangement = 2x and Total number of octahedral voids in this arrangement = x

Order of size :

Triangular void < tetrahedral void

Octahedral void < Cubic void

C.NO.	3	4
	6	8



< → octahedral. voids

In HCP :

unit cell content = 6

No. of octahedral voids = 6

No. of tetrahedral voids = 12 $\boxed{8+12\times\frac{1}{3}}$

In FCC (CCP) :

unit cell content = 4

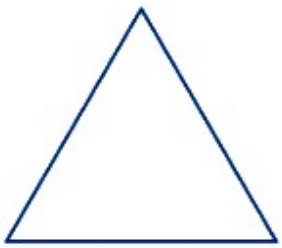
No. of octahedral voids = 4

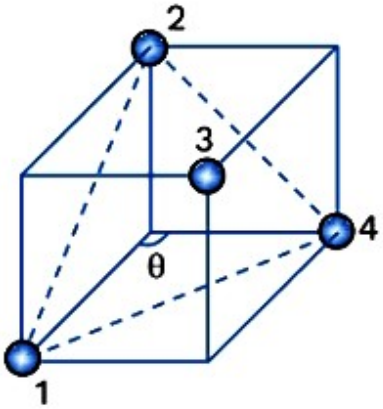
No. of tetrahedral voids = 8

Points to remember

In FCC distance between nearest tetrahedral void & octahedral voids $=\frac{\sqrt{3}}{4}a$

$$1+\left(12\times\frac{1}{4}\right)=1+3=4$$

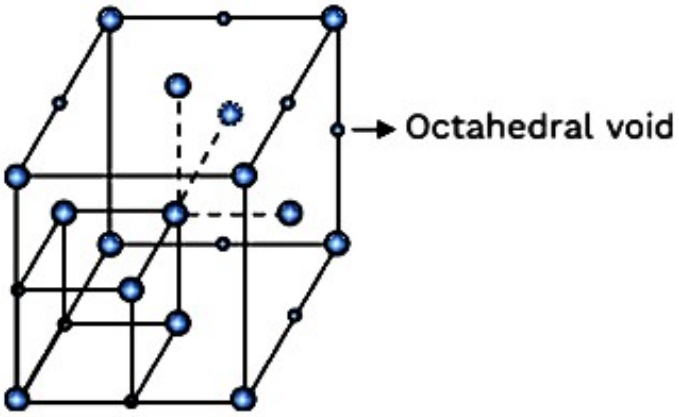




$$\cos\theta=\frac{b^2+c^2-a^2}{2bc}$$

$$= \frac{\left(\frac{b\sqrt{3}}{2}\right)^2 + \left(\frac{b\sqrt{3}}{2}\right)^2 - \left(b\sqrt{2}\right)^2}{2\left(\frac{b\sqrt{3}}{2}\right)\left(\frac{b\sqrt{3}}{2}\right)}$$

$$= 109.3$$



Q. A solid having FCC structure contains X & Y atom.

X-atoms forms FCC lattice

Y-atoms occupy all tetrahedral voids what will be formula of solid?

Solution:

Number of atoms of X = 4

Number of atoms of Y = 8

Ratio X:Y = 4:8 = 1:2

Formula of solid will be

$\therefore XY_2$

Q. A solid consists of X, Y & Z having FCC arrangement

→ occupies all corners

Y → occupies all faces centres

Z → occupies half of the tetrahedral voids

Solution:

formula → $X_{\frac{1}{8}\times 8}\, Y_{\frac{1}{2}\times 6}\, Z_{\frac{8}{2}}$

XY_3Z_4

Q26. **Which of the following statements are correct ?**

(A) An NaCl type AB crystal lattice can be interpreted to be made up of two individual fcc unit cells of A^+ and B^- fused together in such a manner that the corner of one unit cell becomes the edge centre of the other.

(B) In a face centred cubic unit cell, the body centre is an octahedral void.

(C) In fcc unit cell, octahedral and tetrahedral voids are equal.

(D) Tetrahedral voids = 2 × octahedral voids, is true for only ccp and hcp.

Sol. **(A,B,D)**

The number of octahedral voids present in a lattice is equal to the number of close packed particles and the number of tetrahedral voids generated is twice this number.

Q27. **In a compound, atoms of element Y form ccp lattice and those of element X occupy 2/3rd of tetrahedral voids. The formula of the compound will be**

(A) X_3Y_4 **(B) X_4Y_3** **(C) X_2Y_3** **(D) X_2Y**

Sol. **(B)**

Number of Y atoms per unit cell in ccp lattice (N) = 4 Number of tetrahedral voids = 2N = 2 × 4 = 8

Number of tetrahedral voids occupied by X = 2/3rd of the tetrahedral void = 2/3 × 8 = 16/3 Hence, the formula of the compound will be $X_{16/3}Y_4 = X_4Y_3$

Q28. **Element 'B' forms ccp structure and 'A' occupies half of the octahedral voids, while oxygen atoms occupy all the tetrahedral voids. The structure of bimetallic oxide is**

(A) AB_2O_4 **(B) A_4B_2O** **(C) A_2B_2O** **(D) A_2BO_4**

Sol. **(A)**

Number of atoms per unit cell in ccp = N = 4

Number of octahedral voids = N = 4

A occupies half of the octahedral voids, thus A atoms per unit cell = 4/2 = 2 Tetrahedral voids = 2 × N = 8

So number of oxygen atoms = 8

A : B : O :: 2 : 4 : 8

Thus, structure of bimetallic oxide is AB_2O_4.

Q29. **The number of tetrahedral and octahedral voids in a ccp array of 200 atoms are respectively**

(A) 400 and 200 **(B) 200 and 200**

(C) 200 and 400 **(D) 100 and 100**

Q30. **The hcp and ccp structure for a given element would be expected to have**

(A) the same coordination number

(B) the same density

(C) the same packing fraction

(D) same number of atoms in a single unit cell

Sol. **(A,B,C)**

The hcp and ccp structure for a given element would be expected to have:

(A) same coordination number of 12.

(B) same density if made up of the same element as mass and volume occupied and volume available for each atom is same.

(C) same packing fraction. In both unit cells, 74% of total volume is occupied.

(D) different number of atoms in single unit cell.

Structure of Ionic Solids

r_- → radius of anions

r_+ → radius of cations

Ionic solids consist of anions & cation

Anions form the unit cell

And cations occupy voids according to the size

Occupancy of cation	$\frac{r_+}{r_-}$
Triangular void	$0.155 < \frac{r_+}{r_-} < 0.225$
Tetrahedral void	$0.225 < \frac{r_+}{r_-} < 0.414$
Octahedral void	$0.414 < \frac{r_+}{r_-} < 0.732$
Cubic void	$0.732 < \frac{r_+}{r_-} < 1$

Radius ratio	Cations occupy	C.N. of cation	Equation of arrangement
$\frac{r_+}{r_-} < 0.155$	Linear void	2	
$\frac{r_+}{r_-} = 0.155$ limiting	Triangular void	3	$2r_- = b$ $r_+ + r_- = \frac{b}{\sqrt{3}}$ Anion onion contact

$\frac{r_+}{r_-} < 0.225$	Triangular void	3	$r_+ + r_- = \frac{b}{\sqrt{3}}$
$\frac{r_+}{r_-} = 0.255$ limiting	Tetrahedral void	4	In FCC $r_+ + r_- = a_{fcc}\frac{\sqrt{3}}{4}$ $2r_- = \frac{a_{fcc}}{\sqrt{2}}$ anion-anion content
$0.225 < \frac{r_+}{r_-} < 414$	Tetrahedral void	4	In FCC $r_- + r_+ = a_{fcc}^{-}\frac{\sqrt{3}}{4}$ No. anion-anion contact
$\frac{r_+}{r_-} = 0.414$ limiting	Octahedral void	6	In FCC $2r_- = \frac{a_{fcc}}{\sqrt{2}}$ $r_+ + r_- = \frac{a_{fcc}}{2}$ Anion-Anion contact
$0.414 < \frac{r_+}{r_-} < 0.732$	Octahedral void	6	In FCC $r_+ + r_- = \frac{a_{fcc}}{2}$ No. anion-anion contact

			In SC
$\frac{r_+}{r_-} = 0.732$	Cubic void	8	$2r_- = a_{sc}$ $r_+ + r_- = a_{sc}\frac{\sqrt{3}}{2}$ Anion-Anion contact
$0.732 < \frac{r_+}{r_-} < 1$	Cubic void	8	In SC $r_+ + r_- = a_{sc}\frac{\sqrt{3}}{2}$ No. anion-anion contact

Q31. **Identify the correct statements.**
(A) CsCl changes to NaCl structure on heating.
(B) NaCl changes to CsCl structure on applying pressure.
(C) Co-ordination number decreases on applying pressure.
(D) Co-ordination number increases on heating.

Sol. **(A,B)**
Co-ordination number increases on applying high pressure and decreases on heating.

Q32. **A compound XY crystallizes in bcc lattice with unit cell edge length of 280 pm. If the radius of Y^- is 125 pm, then the radius of X^+ (in pm) is__.**

Sol. **(117.2)**

$$r_+ + r_- = \frac{\sqrt{3}}{2}a \text{ or } r_+ + 125 = \frac{\sqrt{3}}{2} \times 280$$

r_+ = 117.2 pm

Q33. **A solid compound XY has NaCl structure. If the radius of the anion is 50 pm, then radius of the cation (in pm) will be_____.**

Sol. **(20.7)**

For NaCl, cationic radius will be equal to the radius of octahedral void.

$\frac{r_+}{r_-} = 0.414$

Given, radius of anion = 50 pm $r_+ = 50 \times 0.414 = 20.7$ pm

Some Ionic solids at a glance

(1) Rock salt structure (AB type)

NaCl, KCl, KBr, LiF, etc.

$$\boxed{\frac{r_{Na^+}}{r_{Cl^-}} \approx 0.53}$$

→ Cl^- ions (anion) form CCP (FCC) while Na^+ ions occupy all the octahedral voids.

→ No of chlorides ion in unit cell is equal to 4

→ No. of sodium ion in one unit cell is = 4

→ formula of one unit cell $\frac{4\text{Mamu}}{\frac{4M}{N_A}\text{gm}}$

→ Density of solid $= \frac{4M}{N_A V}\text{gm/cm}^3$

→ **Equation:**

→ $\boxed{r_+ + r_- = \frac{a_{fcc}}{2}}$ **volume of unit cell :**

$$\boxed{= a_{fcc} = \left[2\left(r_+ + r_-\right)\right]^3} \qquad \text{P.F.} = \frac{\left(x\frac{4}{3}\pi r_+^3\right) + \left(y \times \frac{4}{3}\pi r_-^3\right)}{a_{fcc}^3}$$

→ Coordination No. of Cl^- = 6 Na^+ at a distance $\frac{a_{fcc}}{\sqrt{2}}$

(Next)'neighbour of Cl^- = 12 cl^- at a distance $\frac{a_{fcc}}{\sqrt{2}}$

Points to remember:

$A_x B_y$ formula

$$\frac{x}{y} = \frac{\text{C.N. of B}}{\text{C.N. of A}}$$

e.g. NaCl → 6 : 6 solid

P.F. of ionic solid

A_x	B_y
+	-

Q34. If the radius of Na^+ is 95 pm and that of Cl^- ion is 181 pm then
(A) coordination no. of Na^+ is 6 (B) co-ordination no. of Na^+ is 8
(C) length of the unit cell is 552 pm (D) length of the unit cell is 380 pm.

Sol. (A,C)

$$\frac{r_{Na^+}}{r_{Cl^-}} = \frac{95}{181} = 0.524$$

It lies in the range of 0.414 – 0.732 for which C.N. is 6 and geometry is octahedral.
$a = 2r_{Na^+} + 2r_{Cl^-} = 2 \times 95 + 2 \times 181 = 552$ pm.

Q35. If *AB* crystallises in NaCl structure with *B*$^-$ ions at lattice points, what fraction of octahedral voids is occupied by *A*$^+$ ions?
(A) 25% (B) 50% (C) 100% (D) 75%

Sol. (C)

In NaCl crystal, number of Na^+ ions is equal to Cl^- ions. However, the number of octahedral voids are equal to the number of atoms forming the cubic lattice.
∴ Fraction of octahedral voids occupied by A^+ ion = 100%.

Q36. **Which of the following statements is correct?**

(A) An NaCl type *AB* crystal lattice can be interpreted to be made up of two individual *fcc* unit cells of A^+ and B^- fused together in such a manner that the corner of one unit cell becomes the edge centre of the other.

(B) Octahedral void is combination of two triangular voids surrounded by four spheres.

(C) In *fcc* cubic unit cell size of octahedral and tetrahedral voids are equal.

(D) Number of tetrahedral voids = number of octahedral voids.

Sol. **(A)**

The number of octahedral voids present in a lattice is equal to the number of close packed particles and the number of tetrahedral voids generated is twice of this number. Tetrahedral voids = 2 × octahedral voids.

Octahedral void is combination of two triangular voids surrounded by six spheres.

(2) CsCl structure (AB type) :

CsCl, CsBr, RbCl etc.

→ Cl^- ions, anions form SC. While Cs^+(Cations) occupy cubic voids

→ No. of chlorides ion in one unit cell = 1

→ No of Cs^+ ion in one unit cell is = 1

→ formula of solid $= Cs_1Cl_{\frac{1}{8}\times 8}$

$= CsCl$

→ Mass of one unit cell = 1M amu

$$= \frac{1M}{N_A} gm$$

→ Density of solid $= \frac{1M}{N_A \times V} gm/cm^3$

→ **Equation** $\boxed{r_+ + r_- = \frac{a_{sc}\sqrt{3}}{2}}$

No anion-anion contact

→ Volume of one unit cell $= a_{sc}^3$

→ Coordination No. of Cs^+ = 8 Cl^- at a distance $a_{sc}\frac{\sqrt{3}}{2}$

→ Coordination no. of Cl^- = 8 Cs^+ at distance $\frac{a_{sc}\sqrt{3}}{2}$

(8 : 8 solid)

→ **(Next) :** neighbour of Cs^+ = 6 Cs^+ at a distance a_{sc}

(Next) : neighbour of Cl^- = 6 Cs^+ at distance a_{sc}

Q37. **CsCl crystallises in body-centred cubic lattice. If 'a' is its edge length then which of the following expressions is correct?**

(a) $r_{Cs^+} + r_{Cl^-} = \sqrt{3}a$ **(b)** $r_{Cs^+} + r_{Cl^-} = 3a$

(c) $r_{Cs^+} + r_{Cl^-} = \frac{3a}{2}$ **(d)** $r_{Cs^+} + r_{Cl^-} = \frac{\sqrt{3}}{2}a$

Sol. **(D)**

In a body-centred cubic (*bcc*) lattice, oppositely charged ions touch each other along the cross-diagonal of the cube.

In case of CsCl, $2r_{Cs^+} + 2r_{Cl^-} = \sqrt{3}a$ or $r_{Cs^+} + r_{Cl^-} = \frac{\sqrt{3}}{2}a$

Q38. **In CsCl, Cl^- ion is present at**

(A) body centre **(B) face centre**

(C) edges **(D) at corners of cube.**

Sol. **(D)**

Cl^- is present at corners of cube.

Q39. **Which of the following statements is correct regarding CsCl type of structure of an ionic compound?**

(A) Coordination number of each cation and anion is eight.

(B) Coordination number of each cation and anion is six.

(C) Cation occupies the edge of the unit cell.

(D) Anions are primitive cubic or present at the centre of the unit cell.

Sol. **(A)**

In CsCl, coordination number of each cation and anion is eight.

(3) Fluorite structure (AB_2 type)

CaF_2, BaF_2, SrF_2 etc.

→ Ca^{2+} ions (cations) from CCP (FCC) white F^- ions occupy all the tetrahedral voids

→ No. of F^- ion in one unit cell is = 8

→ No. of Ca^{2+} ions in one unit cell is = 4

→ Formula of solid = Ca_4F_8,

= CaF_2

→ Mass of one unit cell = 4 M amu

$$\frac{4M}{N_A}\text{gm}$$

→ Density of solid $= \frac{4M}{N_A V}\text{gm/cm}^3$

→ Equation $\boxed{r_+ + r_- = a_{fcc}\frac{\sqrt{3}}{4}}$

No. anion-anion contact

→ Volume of one unit cell

$$= \left(\frac{4(r_+ + r_1)}{\sqrt{3}}\right)^3$$

→ C.N. of Ca^{2+} = 8 F^-

Coordination. of F^- ion = 4 Co^{+2}

→ **(Next)** neighbour of Ca^{+2} = $12Ca^{+2}$

(Next)^{1}neighbour of F^- = 6 F^-

(8 : 4 solid)

Q40. **How many of the given compounds belongs to the fluorite type structure?**
CaF_2, SrF_2, BaF_2, PbF_2, $CaCl_2$

Sol. **(5)**
All the given ionic compounds have fluorite type structure.

Q41. **The melting point of RbBr is 682°C, while that of NaF is 988°C. The principle reason that melting point of NaF is much higher than that of RbBr is that**
(A) the two crystals are not isomorphous
(B) the molar mass of NaF is smaller than that of RbBr
(C) the intermolecular distance ($r_c + r_a$) is greater for RbBr than that for NaF
(D) the bond in RbBr has less covalent character than the bond in NaF.

Sol. **(C,D)**
Both option (c) and (d) are correct. Due to large size of RbBr than NaF, intermolecular distance increases. It results in decrease in melting point.

Points to remember

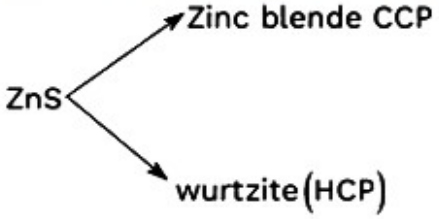
In diamond all carbon cations are p^3 hybridised coordination no. of each carbon atom is 4 that why unit cell of carbon ion be can,.

(4) **Antifluorite structure (B_2A) type**

Na_2O, Li_2O, K_2O, K_2S etc.

→ Oxide ions (anions) form CCF white Na^+ions cations occupy all the tetrahedral voids.

(5) **Zinc bland structure (AB type) :**



$$\frac{r_{Zn^{2+}}}{r_{S^{2-}}} \approx 0.25$$

→ S^{2-}ions (anions) from CCP (FCC) white

Carbons atoms can form CCP or HCP white four alternate tetrahedral voids are occupied by carbon atom itself
Mass of one unit cell

$= 18 \times 12$ amu $= \frac{96}{N_A}$ gm

→ Density of one unit cell

$= \frac{96}{NaV}$ gm / cm^3

Zn^{2+} occupy 1/2 of the tetrahedra voids.

→ No. of S^{2-} ions in one unit cell is = 4

→ No. of Zn^{2+} ions in one unit cell is =4

→ Formula of solid = $Zn_{\frac{1}{2}\times 8}S_4$

$= Zn_4S_4$

$= ZnS$

→ Mass of one unit cell = $\frac{4\times M}{N_A}$ gm

→ Density of one unit cell = $\frac{4M}{N_A V}$ gm / cm^3

→ Equation: $\boxed{r_+ + r_- = a_{fcc}\frac{\sqrt{3}}{4}}$

$a_{fcc} = \frac{4}{\sqrt{3}}(x_+ + x_-)$

→ Volume of one unit cell = $(a_{fcc})^3$

→ Coordination no. of S^{2-}= 4

Coordination no of Zn^{2+} = 42^{+2+}

→ (Next)^1 neighbour of S^{2-} = 12

(Next)^1 neighbour of Zn^{2+} = 12

(4 : 4 solid)

Q42. Which of the following statements are correct for the diamond structure?

(A) Each atom has 4 nearest neighbours.

(B) It is relatively empty.

(C) The maximum proportion of the available volume which may be filled by hard spheres is only 0.34.

(D) The maximum proportion of the available volume which may be filled by hard spheres is only 0.46.

Sol. (A,B,C)

The packing fraction (PF) $= \dfrac{8 \cdot \frac{4}{3}\pi r^3}{a^3}$ Where $2r = \dfrac{\sqrt{3}a}{4} \Rightarrow \dfrac{r}{a} = \dfrac{\sqrt{3}}{8}$

Hence, PF is $\dfrac{32}{3}\pi\left(\dfrac{3\sqrt{3}}{8\times 64}\right) = \dfrac{\sqrt{3}\pi}{16} \approx 0.34$

Q43. **TiO_2 structure is known as**

(A) Rutile structure **(B) Fluorite structure**

(C) Spinel structure **(D) Salt structure**

Sol. (A)

TiO_2 is known as Rutile structure.

Q44. **Which of the following statements are correct?**

(A) A rutile (TiO_2) structure consists of an *hcp* anion lattice with cations occupying half the octahedral holes.

(B) In the Wurtzite structure, the coordination no. of each type of ion is 4.

(C) In the fluorite structure (CaF_2), anions occupy tetrahedral holes in an expanded *fcc* lattice of cations.

(D) A rutile (TiO_2) structure consists of an *hcp* anion lattice with cations occupying all the octahedral holes.

Sol. **(A,B,C)**

A rutile (TiO_2) structure consists of an *hcp* anion lattice with cations occupying half the octahedral holes.

Q45. **Which of the following crystals show 8 : 4 coordination?**

(A) $BaCl_2$ **(B) BaF_2** **(C) SrF_2** **(D) PbF_2**

Sol. **(ABCD)**

All of these show 8 : 4 coordination.

Q46. **The fluorite structure shows 8 : 4 coordination. Which of the following solids have fluorite-like lattices?**
(A) CaF_2 (B) $SrCl_2$ (C) BaF_2 (D) ThO_2

Sol. **(A,B,C)**
CaF_2, $SrCl_2$ and BaF_2 have fluorite like lattices.

INPERFECTIONS IN SOIDS (Crystal Defect):

When some particles are dislocated or missing from their original position then crystal is known as defective crystal.

Thermodynamics explanation of point defect

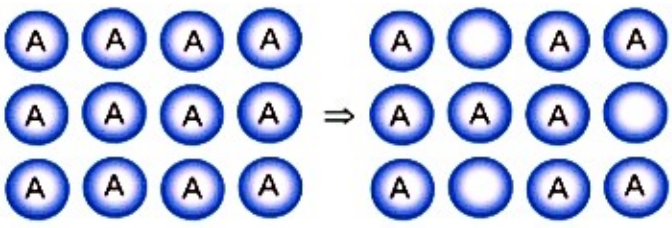
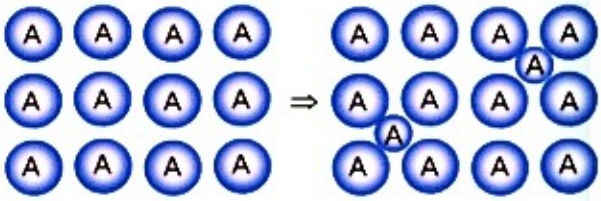
Perfect crystal $\rightleftharpoons$ Imperfect crystal

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$
$$\oplus \quad - \uparrow \oplus$$

On increasing temperature, formation of defects will be favour because it is entropy governing process.

Point defects in metallic solids

Vacancy defects
Interstitial defects

Vacancy defects	**Interstitial defects**
	
Some particles are missing from original position. **Theoretical density in such cases will be higher than actual density.**	Some extra particles will be present at interstitial sites. **Theoretical density in such cases will be lower than actual density.**

Points defects in ionic solids

There are three types of point defects in ionic solids

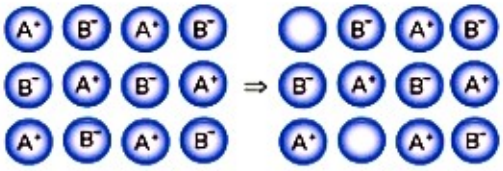
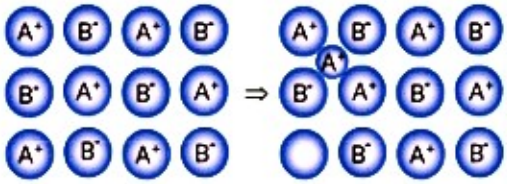
Stoichiometric defects

Impurity defects

Non-stoichiometric solids

Stoichiometric defect	non-stoichiometric defect
Due to such defects ratio of number of anions to cations does not change therefore empirical formula of compound remains same.	Due to these defects the ration number of cations to anions changes. Therefore, empiric formula changes.

(1) Stoichiometric defects in ionic solids

	Schottky defects		Frenkel defects
1.		1.	
2.	Equal number of cations & anions are missing from their positions	2.	Some smaller ions usually cations are dislocated from their original positions to interstitial sites.
3.	It is observed when ionic size of anions & cations are almost same & crystal have high coordination no. like 6 : 6 or 8 : 8 etc	3.	It is observed when ionic sizes of anions & cations are different & solid has low coordination no like 4 : 4, 8 : 4 etc.
4.	E.g. NaCl, CsCl, KCl, AgBr, etc.	4.	Eg. ZnS, AgCl, AgBr, AgI, CaF_2.

5.	Theoretical density will be higher than actual density.	5.	Since ions are not missing rather occupying the interstitial sites in the lattice, theoretical density will be same as experimental density.
6.	Due to this defect ionic solid becomes conducting in nature. They can be p-type and n-type semi-conductor	6.	Due to such defects solids become semi-conductors. Due to small size of cations these have high covalent character and have high dielectric constant.

Q47. **Which of the following statements are incorrect in the context of point defects in a crystal?**

(A) AgCl has anion Frenkel defects and CaF_2 has Schottky defects.

(B) AgCl has cation Frenkel defects and CaF_2 has anion Frenkel defects.

(C) AgCl as well as CaF_2 have anion Frenkel defects.

(D) AgCl as well as CaF_2 have Schottky defects.

Sol. **(A,C,D)**

AgCl has cation Frenkel defects and CaF_2 has anion Frenkel defects.

Q48. **Which of the following statements for crystals having Frenkel defect are correct?**

(A) Frenkel defects are observed where the difference in size of cations and anions is large.

(B) The density of crystals having Frenkel defect is lesser than that of the pure perfect crystals.

(C) In an ionic crystal having Frenkel defect may also contain Schottky defects.

(D) Usually alkali halides do not have Frenkel defects.

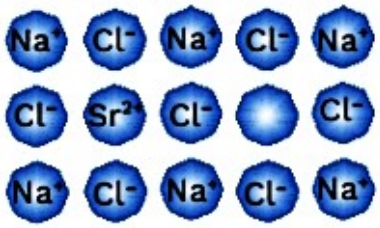
Sol. **(A,C,D)**

Frenkel defects are more common in ionic compounds with low co-ordination number and where there is large difference in size between positive and negative ions. For example, ZnS, AgCl, AgBr and AgI.
In pure alkali metal halides, these defects are not very common because the ions cannot get into interstitial positions due to their large sizes.

Q49. Which of the following compound(s) is/are likely to show both Frenkel and Schottky defects in its crystalline form?
(i) AgBr (ii) CsCl (iii) KBr (iv) ZnS
(A) II and III only (B) III only
(C) III and IV only (D) I only

Sol. (D)
AgBr shows both Frenkel as well as Schottky defects.

(2) **Impurity Defect:**
When concentrated solution of NaCl containing some impurities of $SrCl_2$ is crystalised, $SrCl_2$ is also crystalises out by NaCl. Some positions of NaCl are occupied by $SrCl_2$. Since Sr^{2+} occupies position of Na^+, to maintain neutrality some vacancies are created at the positions which could have been occupied by Na^+. Such defects are called impurity defects.



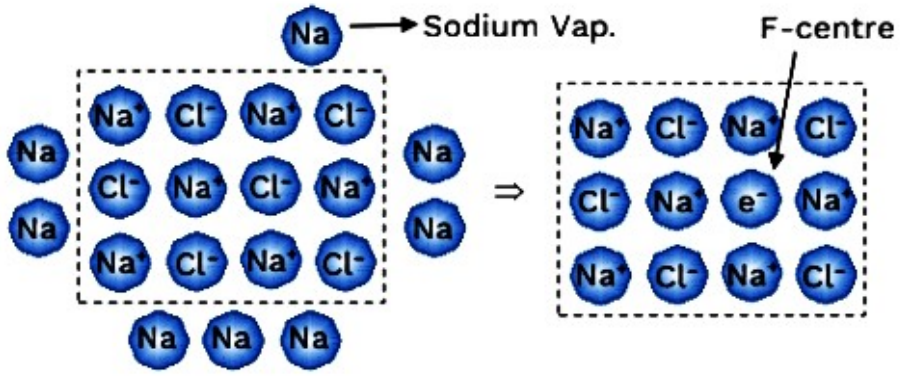
No. of vacancies created = Number of Sr^{2+} ions.

$CdCl_2$ in AgCl is another example of such defect.

(3) **Non-stoichiometric defects**
There are two types of non-stoichiometric defects in ionic solids namely, metal excess defects and metal deficiency defects.

(A) **Metal excess defect** are created due to anionic vacancies. Let's understand such defects by taking example,

When NaCl crystal is heated in presence of excess sodium vapour, electrons released diffuse into the crystal while Cl^- ions will effuse out from crystal. These electrons get trapped in the vacancies created by anions known as F-centre (colour centre). Due to presence of F-centre the crystal becomes paramagnetic, semi-conductor (n-type semi-conductor)



Points to remember

i. Excess of Na make NaCl golden yellow
ii. Excess of K makes KCl purple or linolic.
iii. Excess of Li makes LiCl pink.

iv. $ZnO(s) \rightleftharpoons Zn^{2+} + \frac{1}{2}O_2 \uparrow + 2eF^-centre$

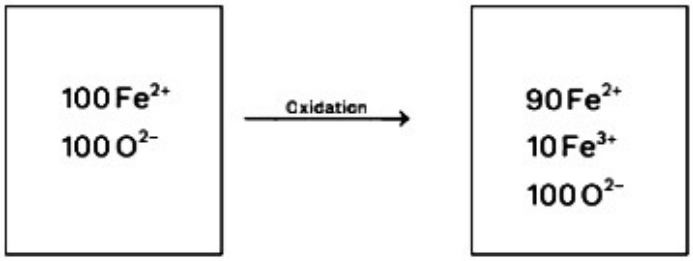
v. PbO, CdO, Cr_2O_3, Fe_2O_3 etc.

(B) Metal deficiency defect is found in ionic solids of transition metals having variable oxidation state:

e.g., non- stoichiometric defect in FeO

Fe_xO $0.84 < x < 0.97$

FeO



$Fe_{100}O_{100}$ $\quad 10Fe^{2+} \rightarrow 10Fe^{3+}$ $\quad Fe_{95}O_{100}$

FeO $\quad Fe_{0.95}O$

Due to partial oxidation of Fe^{2+} to Fe^{3+} positive charge increases to maintain charge neutrality some Fe^{+} & Fe^{+3} will be diffused out of crystal. So total no of Fe decreased. So, it makes non-stoichiometric.

e.q. Cl_8S, $Ni_{1.7}O$, $TiO_{1.44}$ etc.

Q50. Experimentally it was found that a metal oxide has formula $M_{0.98}O$. Metal M, is present as M^{2+} and M^{3+} in its oxide. Fraction of the metal which exists as M^{3+} would be

(A) 5.08% **(B) 7.01%** **(C) 4.08%** **(D) 6.05%**

Sol. (C)

Let the fraction of metal which exists as M^{3+} be x. Then the fraction of metal as M^{2+} = $(0.98 - x)$

$\therefore 3x + 2(0.98 - x) = 2$

$x + 1.96 = 2 \Rightarrow x = 0.04$

$$\therefore \quad \% \text{ of } M^{3+} = \frac{0.04}{0.98} \times 100 = 4.08\%$$

Q52. With which of the following elements silicon should be doped so as to give p-type semiconductor?

(A) Selenium **(B) Boron** **(C) Germanium** **(D) Aluminium**

Sol. (B,D)

If silicon is doped with any of the element of group 13 (B, Al, Ga, In, Tl) of the periodic

table, *p*-type of semiconductor will be obtained.

Q53. **Which of the following are ferroelectric?**
(A) Rochelle salt **(B) $K_4[Fe(CN)_6]$** **(C) $BaTiO_3$** **(D) KH_2PO_4**

Sol. **(A,C,D)**
$K_4[Fe(CN)_6]$ is not a ferroelectric compound.

Q54. **Which of the following are incorrect about Piezoelectric crystals?**
(A) They produce an electric current on heating.
(B) They produce an electric current when a mechanical stress is applied.
(C) They are insulators.
(D) They are magnetic in nature.

Sol. **(ACD)**
Piezoelectricity is production of electricity when crystals having a net dipole moment are subjected to pressure or stress.

Q55. **For a dielectric which of the following properties hold good?**
(A) They are superconductors at high temperatures.
(B) They are superconductors at low temperatures.
(C) They can never become a superconductor.
(D) They have very less dielectric breakdown voltage.

Ans. **(B)**
Superconductors are characterised by diamagnetism behaviour and zero resistivity, which true for a dielectric. They occur only at low temperature. Thus, a dielectric can become a superconductor at low temperatures with very high dielectric breakdown voltage.

Q56. **When does a dielectric become a conductor?**
(A) At avalanche breakdown **(B) At high temperature**
(C) At dielectric breakdown **(D) In the presence of magnetic field**

Sol. **(C)**

When a dielectric is placed in an electric field and if the electric field is increased, when the electric field exceeds the critical field, the dielectric loses its insulating property and becomes conducting. This is called dielectric breakdown.

Q57. **Which among the following compounds can show the properties of a Ferroelectric substance?**

(A) $BaTiO_3$ **(B) $PbZrO_3$** **(C) MnO_2** **(D) CrO_2**

Sol. **(A)**

At the curie temperature (around 120°C) paraelectric $BaTiO_3$ transforms into a ferroelectric structure. Ferroelectricity is the characteristic of certain substances that possess a spontaneous electric field which can be reversed by applying an external electric field.

Q58. **Which among the following statements is correct?**

(A) NaCl is a paramagnetic substance.

(B) Paramagnetic substances behave like an insulator.

(C) Cobalt is an antiferromagnetic substance.

(D) On heating, ferrimagnetic substances lose ferrimagnetism.

Sol. **(D)**

On heating, ferrimagnetic substances become paramagnetic and hence, lose their ferrimagnetism. NaCl is diamagnetic in nature. Cobalt is a ferromagnetic substance.

Q59. **Which of the following compounds is metallic and ferromagnetic?**

(A) TiO_2 **(B) CrO_2** **(C) VO_2** **(D) MnO_2**

Sol. **(B)**

CrO_2 is metallic and ferromagnetic

Q60. **Which of the following statements is incorrect about semiconductors?**

(A) Impurity of lower group creates *p*-type semiconductors.

(B) Impurity of higher group creates *n*-type semiconductors.

(C) Intrinsic semiconductors are formed by doping impurity.

(D) Intrinsic semiconductors become conductors when temperature is raised.

Sol. **(C)**

Intrinsic semiconductors are pure semiconductor and become conductors when temperature is raised.

Q61. **Group 14 is doped with group 15 create which type of semiconductor?**

(A) p-type **(B) n-type** **(C) n, p-type** **(D) Intrinsic**

Sol. **(B)**

14th group doped with 15th group forms *n*-type semiconductor.

Q62. **Which one of the following statements is wrong?**

(A) The conductivity of metals decreases with increase in temperature.

(B) The conductivity of semiconductors increases with increase in temperature.

(C) There is no superconductor at room temperature.

(D) Ionic solids conduct electricity due to presence of ions.

Sol. **(D)**

Ionic solids conduct electricity not due to presence of ions but due to presence of defects.

Q63. **'X' melts at low temperature and is a bad conductor of electricity in both liquid and solid state. X is**

(A) carbon tetrachloride **(B) silicon carbide**

(C) mercury **(D) zinc sulphide.**

Sol. **(A)**

CCl_4 is a non-conductor in both solid and liquid state. CCl_4 is covalently bonded molecule, so the electrons cannot flow and there are no ions. Hence, it does not conduct electricity in any state.

Magnetic properties of solids

Though, magnetic properties of solids mainly arise from electronic bands which will be discussed in physics, here we will pay attention to the magnetic properties arising from arrangement of electrons, their orbital and spin magnetic moments.

Magnetic properties can be classified into several categories as follow:

Diamagnetic

Substances which are weakly repelled in presence of a magnetic field. Electrons in orbitals of such substances are fully paired. For instance, sodium chloride (NaCl) in solid state is diamagnetic. These substances are mainly insulators of electricity.

Paramagnetic

Such substances are weakly attracted by magnetic field. In absence of magnetic field their magnetism is lost. For a substance to show behave as paramagnetic it must have at least one unpaired electron. They are conductors of electricity and used in electronic appliances. Dioxygen(O_2), Cr^{3+}, Fe^{3+} are some examples of paramagnetic substances.

Ferromagnetic

They are strongly attracted by magnetic field and have all unpaired electrons are arranged parallelly. One removal of magnetic field if the alignment of electrons remains parallel then substance is said to be permanently magnetised. Iron, cobalt, CrO_2 etc are some ex-

amples of Ferromagnetic substances.

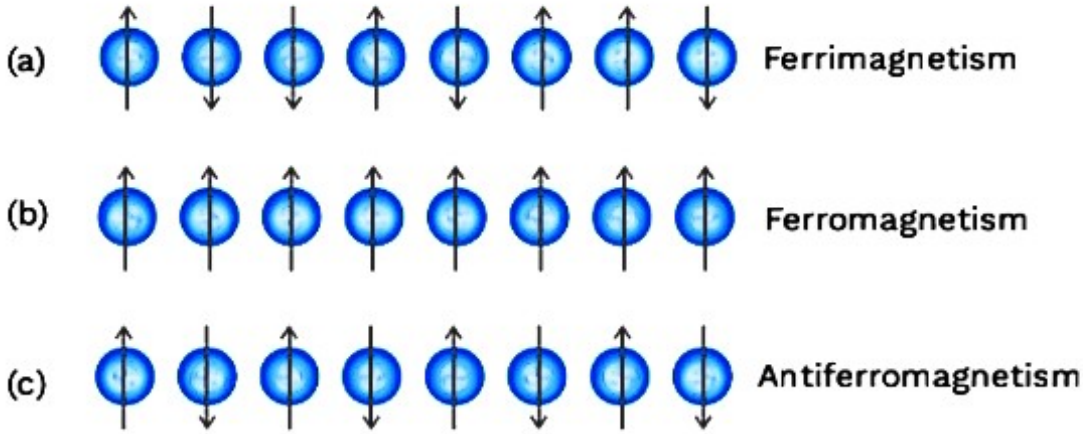
Antiferromagnetic

Unpaired electrons in such species are arranged in anti-parallel directions, due to which their net magnetic moment comes out to be zero even though they have unpaired electrons.

Some examples of such substances are NiO, MnO etc.

Ferrimagnetism

Arrangement of electrons is in such a way that parallel and antiparallel orientations don't have a fixed pattern; such substances are called Ferrimagnetic substance. They are strongly attracted by magnetic field as compared to paramagnetic substances but weakly attracted as compared to Ferromagnetic substances. Some examples of such substances are Fe_2O_3, $MgFe_2O_4$ and $ZnFe_2O_4$.



QUICK OVERVIEW

1. $H_2O(S)$ is example of molecular solid.
2. Metallic solids show electrical conduction.
3. Network solids are very hard.
4. Diamond and graphite are examples of network covalent solids.
5. Ionic solids are hard and Brittle and insulator in solid state.
6. Crystalline solids have sharp melting point.
7. Amorphous solids have short range of order of particles.
8. Amorphous solids have range melting point.
9. Crystalline solids are Anisotropic in nature.
10. In two dimension five types of lattice are possible.
11. In two dimension only simple (primitive) and centered unit cell is possible.
12. In three dimension seven crystal classes, four types unit cells and 14 types of lat-

tice are possible.

13. In three Dimension each crystal class do not have all tour different type of unit cell.
14. Face centred unit cell exist only in orthombic crystal class.
15. Cubic crystal only have three (S, BC, FC) unit cells.
16. Square packing in two dimension have packing fraction 0.78.
17. Hexagonal close packing in two-dimension fraction of area occupied by spheres is 90.64%.
18. Simple cubic arrangement have packing efficiency 52%
19. Coordination number in simple cubic packing is 6.
20. Number of next nearest neighbors in simple cubic packing is 12.
21. In simple cubic packing next nearest neighbors are at distance $a\sqrt{2}$.
22. In simple cubic packing next to next nearest neighbors are at distance $a\sqrt{3}$ and they are 8 in number.
23. In body center cubic structure $\sqrt{3}a = 4r$
24. Packing fraction for BCC is 68%.
25. Coordination number for BCC is 8
26. In BCC next nearest neighbors are 6 are they are at distance a.
27. In BCC next to next nearest neighbors are 12 and they are at a distance a $a\sqrt{2}$
28. In BCC next to next to next nearest neighbors are 8 and they are at distance $a\sqrt{3}$
29. In hexagonal unit cell the value of C (height) is $\sqrt{\frac{2}{3}}\ 4r$
30. Volume of HCP unit cell is $24\sqrt{2}\,r^3$
31. In HCP effective number of atoms are 6.
32. Packing fraction of HCP unit cell is 74%.
33. In HCP unit cell coordination number is 12.
34. In CCP (or FCC) plans are arranged along the body diagonal of cubic unit cell.
35. In CCP effective number of atom = 4
36. In CCP coordination number is 12.
37. In CCP there are 12 nearest neighbors and they are at distance $\frac{a}{\sqrt{2}}$
38. In CCP there are 6 next nearest neighbors at a distance a.

39. In CCP there are 24 next to next nearest neighbors at a distance of a $\sqrt{\frac{3}{2}}$.

40. In CCP there are 12 next to next to next nearest neighbors at a distance of $a\sqrt{2}$

41. For tetrahedral void r = 0.225 R and CN = 4.

42. For tringular void r = O.155 R and CN = 3.

43. For octahedral void r = 0.414 R and CN = 6.

44. For cubical void r = 0.732 R, and CN = 8.

45. The FCC unit cell have eight tetrahedral void below each corner of the unit cell.

46. In FCC unit cell effectively four octahedral voids exist.

47. In hCP tetrahedral voids are 12.

48. In hcp octahedral voids are 6.

49. In BCC symmetrical voids are not found.

50. In Ionic solids location of cation and anion depends on the radius ratio (r_+/r_-).

51. If radius ratio is 0.155 then cation is perfectly in triangular void and a = 2r.

52. If $0.155 < \frac{r_+}{r_-} < 0.225$ then anions are not in contact and a = $\sqrt{3}$ (r. + r).

53. If $\frac{r_+}{r_-}$ = 0.225 cation is perfectly in tetrahedral void and $\sqrt{2}\,a_{fcc} = 4r$.

54. If $0.225 < \frac{r_+}{r_-} < 0.414$, FCC lattice will expand and anions are not contact here

$\sqrt{2}\,a_{fcc} \neq 4r$ but $a_{fcc} = \frac{4}{\sqrt{3}}(r_+ + r_-)$.

55. If $\frac{r_+}{r_-}$ = 0.414 cation is perfectly in octahedral void and for FCC lattice $\sqrt{2}a_{fcc} = 4r_-$ valid.

56. If $0.414 < \frac{r_+}{r_-} < 0.732$, FCC lattice undergoes expansion and anions loses contact having $a_{fcc} = 2(r_+ + r_-)$.

57. If $\frac{r_+}{r_-}$ = 0.732 cation is perfectly in cubical coid and $a_{SC} = 2r$.

58. If $0.732 < \frac{r_+}{r_-} < 1$ anion are not is contact and = $\sqrt{3}a_{fcc} = 2(r_+ + r_-)$

59. In rock salt structure (NaCl type) Cl^- make fcc lattice and Na^+ ion occupy octahedra

void.

60. For NaCl type structure $a_{fcc} = 2 + (r_{Na^+} + r_{cl^-})$
61. In ZnS type structure, anion (S^{2-}) forms fcc lattice and cation (Zn^{2+}) occupy alternate tetrahedral void.

62. Fro ZnS type structure $a_{fcc} = \frac{4}{\sqrt{3}}(r_{Zn^{2+}} + r_{S^{2+}})$

63. In CaF_2 (Fluorite structure) type solids. FCC lattice positions are occupied by Ca^{2+} and F^- aions are at all tetrahedral voids.

64. For fluorite type structure $a_{fcc} = (r_{Ca^{2+}} + r_{F^-}) \times \frac{4}{\sqrt{3}}$
65. Na_2O have antifluorite structure.
66. In antifluorite structure anions are at FCC lattice position and cations are at all tetrahedral voids.
67. CsCl type structure, anions form FCC lattice and cation is at cubical void.

68. For CsCl type structure $a_{sc} = \frac{2}{\sqrt{3}}(r_+ + r_-)$

69. In Schottky defect formula remains same but density get decreased.
70. In Frenkel defect formula and density remains same.

Answer key

1. T	**2. T**	**3. T**	**4. T**
5. T	**6. T**	**7. T**	**8. T**
9. T	**10. T**	**11 T**	**12. T**
13. T	**14. F**	**15. T**	**16. T**
17. T	**18. T**	**19. T**	**20. T**
21. T	**22. T**	**23. T**	**24. T**
25. T	**26. T**	**27. T**	**28. T**
29. T	**30. T**	**31. T**	**32. T**
33. T	**34. T**	**35. T**	**36. T**
37. T	**38. T**	**39. T**	**40. T**
41. T	**42. T**	**43. T**	**44. T**

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